A uniform bound of five for Diophantine tuples in imaginary quadratic integer rings

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Abstract

We prove that every $D(1)$-tuple in the full ring of integers of an imaginary quadratic field has at most five elements. The proof combines determinant estimates, interpolation of square roots, and explicit simultaneous approximation. The remaining cases are reduced to finite computations, for which complete search bounds and output tables are included.

1 Introduction and conventions

Let K be an imaginary quadratic field with full ring of integers $\mathcal{O}_K$. A finite set $\mathcal{A} \subset \mathcal{O}_K \setminus \{0\}$ is a $D(1)$-tuple if $xy+1$ is a square in $\mathcal{O}_K$ whenever $x, y \in \mathcal{A}$ are distinct. Zero is allowed as such a square. Equivalently, the square roots may be required to lie in K. A square root in K of an algebraic integer is again in $\mathcal{O}_K$.

For a general account of Diophantine tuples and their connections with elliptic curves, see Dujella [5].

Adžaga obtained the uniform bound 42 for these tuples [1]. Cipu and Fujita proved the bound 24 for both $D(1)$-tuples and $D(-1)$-tuples in imaginary quadratic integer rings [3]. We obtain the following bound for $D(1)$-tuples.

Theorem 1.1. *For every imaginary quadratic field K, every $D(1)$-tuple in $\mathcal{O}_K$ has cardinality at most five.*

We order a hypothetical sextuple by modulus and distinguish the cases of zero, one, or at least two entries of modulus less than three. Determinant estimates and interpolation give lower bounds for the larger entries. We then construct integral linear forms in square roots that exclude overlapping ranges for the largest entry. Together these estimates settle the case with no small entries and reduce the other cases to bounded computations. For two small entries, a descent generates all common extensions below the required bound.

The simultaneous approximation construction follows Bennett [2] and Jadrijević and Ziegler [9]. We give the polynomial formulas and the coefficient and remainder estimates needed here. The determinant identity of Bennett is stated explicitly. The nonvanishing argument for a 3×3 matrix of roots of $a_i b_j + 1$ also appears in Wildner Fraga's preprint [10, Lemma 3.1]. We include that argument, without using the quantitative Subspace Theorem considered there. The factorization used for the pair $\{-1, 1\}$ also appears in Cipu's lecture slides [4, pp. 27–30]. Franušić [7, Theorem 1] proved that $\{1, 3\}$ cannot be extended to a Diophantine quintuple in $\mathbb{Z}[\sqrt{-d}]$ for positive squarefree $d \neq 2$. Franušić and Kreso [8, Corollary 1.2] treated $d = 2$. For $d \equiv 3 \pmod 4$, the ring $\mathbb{Z}[\sqrt{-d}]$ is properly contained in the full ring of integers of $\mathbb{Q}(\sqrt{-d})$. We handle the pair in the full ring of integers using the same approximation estimates as for the other pairs.

Dujella [6, Problem 1.11] asks whether a $D(1)$-quintuple exists in the ring of integers of an imaginary quadratic field. Our bound does not settle this question.

Fix one complex embedding of K. We write $N(z) = z\overline{z} = |z|^2$. For $z \in \mathcal{O}_K \setminus \{0\}$ this is a positive integer, so $|z| \geq 1$. When lower-case letters denote ordered tuple entries, the corresponding capitals denote their moduli. Equal moduli are allowed throughout. Roots of the elements $xy + 1$ supplied by the tuple are chosen in $\mathcal{O}_K$. Auxiliary square roots used in analytic estimates may lie in $\mathbb{C}$.

For the exact enumerations, write $K = \mathbb{Q}(\sqrt{-d})$, with $d > 0$ squarefree, and

$$\omega = \begin{cases} (1 + \sqrt{-d})/2, & d \equiv 3 \pmod 4, \\ \sqrt{-d}, & d \not\equiv 3 \pmod 4. \end{cases}$$

Then $\mathcal{O}_K = \mathbb{Z}[\omega]$ and $\omega^2 - \tau\omega + \nu = 0$, where $(\tau, \nu) = (1, (d+1)/4)$ or $(0, d)$, respectively. The absolute field discriminant is $\Delta = 4\nu - \tau^2$. Tables using the coordinates $(u + v\sqrt{-d})/2$ are identified explicitly. All other coordinate pairs denote $u + v\omega$. Both descriptions include the full ring of integers.

Simultaneous negation and complex conjugation preserve the tuple property. These, together with permutation, are the only equivalences used in the finite reductions. Multiplication by an arbitrary unit is not used as a tuple symmetry.

All numerical verification below reduces to integer or rational arithmetic. Where an irrational modulus occurs, its square is an explicit integer or rational number and the chosen enclosure is certified by squaring. The finite-sum formulas, bounded algorithms and complete output tables are included in the article.

2 Structural growth estimates

A triple is *regular* if

$$a^2 + b^2 + c^2 - 2ab - 2ac - 2bc = 4,$$

equivalently $c = a + b \pm 2\sqrt{ab + 1}$. This condition is symmetric.

Lemma 2.1. *An irregular triple with $3 \leq A = |a| \leq B = |b| \leq C = |c|$ satisfies $C > 2AB$.*

Proof. Choose $r^2 = ab+1, s^2 = ac+1, t^2 = bc+1$ in $\mathcal{O}_K$, and put $u, v = a+b+c+2abc \pm 2rst$. Their product is $a^2 + b^2 + c^2 - 2ab - 2ac - 2bc - 4$, so irregularity makes both nonzero. Since u, v are nonzero algebraic integers, $|u| + |v| \leq |uv| + 1$. Hence

$$0 \leq C^2 - 2\Big(|1 + 2ab| - |a + b|\Big)C + |(a - b)^2 - 4| + 2|a + b| + 1. \tag{2.1}$$

Using $|1 + 2ab| \geq 2AB - 1$, $|a + b| \leq A + B$, and $|(a - b)^2 - 4| \leq (A + B)^2 + 4$, we obtain $q(C) \geq 0$, where $q(X) = (A + B + X + 1)^2 + 4 - 4ABX$. But

$$\frac{q(B)}{B^2} \leq \left(\frac{10}{3}\right)^2 + \frac{4}{9} - 12 = -\frac{4}{9}.$$

Writing $h = (A + B + 1)/(AB) \leq 7/9$, we also have

$$\frac{q(2AB)}{4A^2B^2} = -1 + h + \frac{h^2}{4} + \frac{1}{A^2B^2} \leq -\frac{19}{324}.$$

Convexity excludes $B \leq C \leq 2AB$. □

Lemma 2.2. *If an ordered quadruple has $1 \le A \le B \le C \le D$ and $B \ge 3$, its last three entries form an irregular triple. In particular $D > 2BC$.*

Proof. Suppose instead $d = b + c + 2r$, $r^2 = bc + 1$. First let $C \ge 4$. Choose $\beta^2 = b, \gamma^2 = c$ so that $r = \beta\gamma\sqrt{1 + 1/(bc)}$, where $\sqrt{1+z}$ is the analytic branch with value 1 at $z = 0$. Since $BC \ge 12$,

$$|r - \beta\gamma| < \frac{20}{39\sqrt{BC}}.$$

Put $p = \beta + \gamma$. The ordering $D \ge C$ gives

$$|p|^2 > C - \frac{40}{39\sqrt{BC}} > \frac{12}{13}C > \left(\frac{24}{25}\right)^2 C.$$

Indeed $\sqrt{BC} > 24/7$, so the relative loss is below $35/468 < 1/13$. Choose $\alpha^2 = a$, and select the integral roots $u^2 = ab + 1$, $v^2 = ac + 1$ using the same analytic branch. Then

$$|u - \alpha\beta| < \frac{5}{9\sqrt{AB}}, \qquad |v - \alpha\gamma| < \frac{7}{13\sqrt{AC}}.$$

For $\delta = ad + 1 - ap^2 = 1 + 2a(r - \beta\gamma)$,

$$\left|\frac{\delta}{ap^2}\right| < \frac{1 + 35/117}{48/13} = \frac{19}{54} < \frac{9}{25}.$$

Consequently the integral root $w = \alpha p\sqrt{1 + \delta/(ap^2)}$ satisfies

$$|w - \alpha p| < \frac{125}{216}\left(\frac{1}{\sqrt{AC}} + \frac{40\sqrt{A}}{39\sqrt{B\,C}}\right).$$

Write $q_0 = 7/13 + 125/216$, $r_0 = 625/1053$. It follows that

$$|u + v - w| < \frac{5}{9\sqrt{AB}} + \frac{q_0}{\sqrt{AC}} + \frac{r_0\sqrt{A}}{\sqrt{B\,C}} < 1.$$

For completeness, the last expression is convex in $\sqrt{A}$ on $[1, \sqrt{B}]$. At the two endpoints, using $B \ge 3, C \ge 4$, its upper bounds are respectively

$$\frac{35}{108} + \frac{q_0}{2} + \frac{7r_0}{48} = \frac{12247}{12636} < 1, \qquad \frac{5}{27} + \frac{7q_0}{24} + \frac{r_0}{4} = \frac{14813}{22464} < 1.$$

Integrality gives $w = u + v$. Squaring and using $d = b + c + 2r$ yields $2(ar - uv) = 1$, impossible because $ar - uv$ is an algebraic integer.

It remains to handle $3 \le B \le C < 4$. The following finite computation is included explicitly. Write $\omega^2 = t\omega - h$ for the usual integral basis of $\mathbb{Q}(\sqrt{-q})$, and $\Delta = 4h - t^2$. Then

$$N(x + y\omega) = x^2 + txy + hy^2, \qquad 4N(x + y\omega) = (2x + ty)^2 + \Delta y^2. \tag{2.2}$$

If either b, c is nonreal, $\Delta \le 60$. If both are rational, they must be $-3, 3$, and the square condition restricts the field to $\mathbb{Q}(\sqrt{-2})$. It suffices to enumerate the following squarefree values of q with $\Delta \le 64$.

$$1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 19, 23, 31, 35, 39, 43, 47, 51, 55, 59.$$

For each field enumerate, by (2.2), all elements of norm at most 15. For distinct b, c with $9 \le N(b) \le N(c) \le 15$, retain $bc + 1 = r^2$, form $d = b + c \pm 2r$, and retain $d \notin \{0, b, c\}$, $N(d) \ge N(c)$. Finally enumerate $a \notin \{0, b, c, d\}$, $N(a) \le N(b)$, testing $ab+1, ac+1, ad+1$. Every square test uses an exact finite table. Since $|d| < 8+2\sqrt{17} < 17$, all roots tested have norm at most 70. Enumerating those roots by (2.2) and squaring them gives the complete square table, including zero.

There are 90 admissible unordered pairs b, c, 118 retained triples, and 16 candidates before the final square test. Simultaneous negation, conjugation, and interchange of b, c reduce these to the six rows below. Coordinates mean $x + y\omega$. Here $\omega = \sqrt{-q}$ unless $q \equiv 3 \pmod 4$, when $\omega = (1 + \sqrt{-q})/2$.

q	a	b	c	d	$ad+1$
2	$(-1,0)$	$(-3,0)$	$(3,0)$	$(0,-4)$	$(1,4)$
2	$(-1,0)$	$(2,-2)$	$(3,0)$	$(11,-4)$	$(-10,4)$
3	$(-1,2)$	$(-1,-3)$	$(4,-3)$	$(7,-14)$	$(22,0)$
6	$(-2,0)$	$(3,-1)$	$(3,1)$	$(14,0)$	$(-27,0)$
11	$(-1,0)$	$(3,1)$	$(4,-1)$	$(15,0)$	$(-14,0)$
35	$(-3,0)$	$(2,1)$	$(3,-1)$	$(13,0)$	$(-38,0)$

The first two final values have nonsquare norms 33 and 132. The third is the positive rational nonsquare 22. For the last three, the ratios to $-q$ are $9/2, 14/11, 38/35$, none a rational square. Thus no row passes the last test. This proves irregularity. The growth assertion follows from Lemma 2.1. □

2.1 The determinant estimates

For entries $a_1, \dots, a_n$, choose $u_i^2 = a_i$ in $\mathbb{C}$ and signs of the integral roots so that

$$M_{ii} = a_i, \quad M_{ij} = u_i u_j \sqrt{1 + 1/(a_i a_j)} \quad (i \ne j).$$

All products below have modulus at least 3, so these branches are defined. Write I for the identity matrix and J for the matrix whose entries are all one, in the required dimension. With $v = (a_i)^T$,

$$N = \operatorname{diag}(u_i) M \operatorname{diag}(u_i) = vv^T + F, \quad F_{ii} = 0, \quad F_{ij} = \frac{1}{1 + \sqrt{1 + 1/(a_i a_j)}}. \tag{2.3}$$

In particular $\det N = (\prod_i a_i) \det M$, and

$$F_{ij} - \frac{1}{2} = -\frac{F_{ij}^2}{2a_i a_j}. \tag{2.4}$$

In even dimension n, M is nonsingular. Indeed, in $\mathcal{O}_K/2\mathcal{O}_K$, entrywise squaring gives

$$(\det M)^2 = \det(vv^T + I + J) = 1.$$

Here $J^2 = 0$, $(I + J)^{-1} = I + J$, $\det(I + J) = 1$, and $v^T(I + J)v = \sum a_i^2 + (\sum a_i)^2 = 0$. The rank-one determinant identity is valid over this quotient ring even when it has zero divisors.

Lemma 2.3. *An ordered quadruple satisfies*

$$D > 2ABC \quad \text{if } A \ge 3, \qquad D > \frac{3}{8}ABC \quad \text{if } A \ge 1,\ B \ge 3.$$

Proof. In the first case Lemma 2.2 gives $D > 6C$ and $D > 18$. For $|z| \le 1/9$, $\operatorname{Re}\sqrt{1+z} > \frac{15}{16}$, so (2.3)–(2.4) give

$$|F_{ij}| < 16/31, \qquad |F_{ij} - 1/2| < e := 128/8649.$$

Put $s = 1/2 + 3e$, $h = 3/2 + 3e$. The matrix $F_0 = (J_4 - I_4)/2$ has singular values $3/2, 1/2, 1/2, 1/2$, and $\|F - F_0\|_2 < 3e$. Therefore

$$|\det F| < hs^3, \qquad \|\operatorname{adj} F\|_2 < hs^2.$$

The latter follows from products of three singular values, including singular matrices by continuity. The rank-one identity yields

$$|\det N| < hs^2\left(s + \sum A_i^2\right) < hs^2\left(1 + \frac{3}{36} + \frac{s}{324}\right)D^2 < \frac{D^2}{2}.$$

The last inequality follows by substituting the definitions of s and h. Since $1 \le |\det M| = |\det N|/(ABCD)$, this proves the first assertion.

For the second assertion, the three off-diagonal entries of $E = F - F_0$ in the row corresponding to a have modulus less than $1/18$, since $|F_{1j}| < 5/9$ and $25/486 < 1/18$. For $2 \le i < j \le 4$, we have $|E_{ij}| < 1/64$. The symmetric nonnegative matrix of these entrywise upper bounds has weighted row sums $1/9, 11/96$ for weights $(3/2, 1, 1, 1)$. Its spectral radius bounds $\|E\|_2$, so $\|E\|_2 < 1/8$. Thus

$$|\det F| < 1625/4096, \qquad \|\operatorname{adj} F\|_2 < 325/512.$$

As $D \ge 3$,

$$|\det N| < \frac{325}{512}\left(\frac{5}{8} + \sum A_i^2\right) \le \frac{95225}{36864}D^2 < \frac{8}{3}D^2,$$

which proves the second assertion by the same integral determinant bound. □

Lemma 2.4. *Five ordered entries of moduli $1 \le A \le B \le S \le C \le D$, with $B \ge 3$, satisfy $D > 3ABSC$.*

Proof. The four large entries give $D > 2BSC \ge 18C$, $D > 54$. In (2.3) take $n = 5$, $w = (a, b, s, c)^T$, $f = (F_{15}, \dots, F_{45})^T$, and let F_{pre} be the submatrix formed by the first four rows and columns of F. The last diagonal entry of N is d^2, so

$$\det N = d^2 \det G, \qquad G = F_{\mathrm{pre}} - \frac{wf^T + fw^T}{d} - \frac{ff^T}{d^2}. \tag{2.5}$$

The weighted bound just proved gives $\|F_{\mathrm{pre}} - (J_4 - I_4)/2\|_2 < 1/8$, hence $\|F_{\mathrm{pre}}^{-1}\|_2 < 8/3$. The last-column products have modulus above 54, so (2.4) gives $|F_{i5} - 1/2| < 625/264627 < 1/400$. For $f_0 = (1/2, \dots, 1/2)^T$,

$$\|f - f_0\|_2 < 1/200, \quad \|f\|_2 < 201/200, \quad \|F_{\mathrm{pre}}^{-1} f\|_2 < \frac{2}{3} + \frac{8}{3}\left(\frac{1}{200} + \frac{1}{12}\right) = \frac{203}{225} < \frac{11}{12}. \tag{2.6}$$

The inverse estimate follows by applying the resolvent identity to $F_0^{-1} f_0 = (1/3, \dots, 1/3)^T$.

For a zero-diagonal symmetric matrix of order four, set $p = F_{12}F_{34}$, $q = F_{13}F_{24}$, $r = F_{14}F_{23}$. Then

$$\det F_{\mathrm{pre}} = p^2 + q^2 + r^2 - 2pq - 2pr - 2qr.$$

Each of p, q, r is within $7/192$ of $1/4$. Expansion gives

$$|\det F_{\mathrm{pre}}| < \frac{3}{16} + \frac{3}{2}\frac{7}{192} + 9\left(\frac{7}{192}\right)^2 < \frac{33}{128}.$$

Put $h = f^T F_{\mathrm{pre}}^{-1} w$, $p_0 = f^T F_{\mathrm{pre}}^{-1} f$, $q_0 = w^T F_{\mathrm{pre}}^{-1} w$. From (2.6),

$$|h| < 11C/6, \qquad |p_0| < 1, \qquad |q_0| < 32C^2/3.$$

The rank-two determinant identity in (2.5) is exactly

$$\det G = \det F_{\mathrm{pre}} \left(1 - \frac{2h}{d} + \frac{h^2 - p_0 q_0 - p_0}{d^2}\right).$$

It follows that

$$|\det G| < \frac{33}{128}\left(1 + \frac{11}{54} + \frac{505}{11664} + \frac{1}{2916}\right) < \frac{1}{3}.$$

Moreover

$$\|G - F_0\|_2 < \frac{1}{8} + \frac{67}{300} + \frac{(201/200)^2}{2916} < \frac{1}{2},$$

so G, and hence M, is nonsingular. Integrality now gives

$$1 \le |\det M| = \frac{D|\det G|}{ABSC} < \frac{D}{3ABSC}.$$

□

2.2 Fourth-entry gaps

Lemma 2.5. *Let an ordered quadruple have $1 \le A \le B \le S \le T$, with $B \ge 3$. If $A \ge 3$, then $T > 3ABS$. If $A < 3$, the first two columns of each row in the following table imply $T > tABS$. The number α is a rational lower bound for A, used in the proof.*

condition on $N(a)$	$B \ge B_0$	α	t
1	8	1	27/8
2	4	140/99	3
3	3	12/7	11/4
4	3	2	29/10
$5 \le N(a) \le 8$	3	20/9	3

Proof. The 4×4 integral matrix constructed above is nonsingular by the same congruence argument. We now sharpen the determinant estimate. Write

$$M = \mathrm{diag}(u_i)(J + E)\,\mathrm{diag}(u_i), \quad E_{ii} = 0, \quad E_{ij} = \delta_{ij} = \sqrt{1 + z_{ij}} - 1, \quad z_{ij} = \frac{1}{a_i a_j}.$$

With triangles and paths taken in the complete graph on the four indices, the cubic part of $\det(J + E)$ is

$$\mathcal{C}(\delta) = 2 \sum_{\text{triangles}} \prod_{\text{edges}} \delta - 2 \sum_{\text{paths}} \prod_{\text{edges}} \delta + 2 \sum_{\{i,j\}} \delta_{ij}^2 \delta_{kl}, \tag{2.7}$$

where the middle sum runs over the twelve undirected Hamiltonian paths, and $\{k, l\}$ is complementary to $\{i, j\}$ in the last sum. Its quartic part is $\det E$, the sum of the squares of

$\delta_{12}\delta_{34}$, $\delta_{13}\delta_{24}$, and $\delta_{14}\delta_{23}$, minus twice the sum of their pairwise products. These identities follow by the 24 permutations of the determinant. Substituting $\delta = z/2$ in (2.7) gives

$$abst\,\mathcal{C}(z/2) = \frac{1}{4}\sum_i \frac{a_i}{\prod_{j\neq i} a_j} - \frac{1}{4}\sum_{i<j} \frac{1}{a_i a_j}. \tag{2.8}$$

Indeed, each triangle gives a term $a_i/\prod_{j\neq i} a_j$ with coefficient $1/4$. For each term $1/(a_i a_j)$, two paths and one squared-edge term contribute.

For $0 \le \rho \le 1/5$, put $k(\rho) = 1/2 + \rho/7$. Rationalization and the identity $\delta - z/2 = -\delta^2/2$ imply

$$|\delta| \le k(\rho)|z|, \qquad |\delta - z/2| \le (\rho/7)|z| \quad (|z| \le \rho). \tag{2.9}$$

To check both bounds explicitly, use

$$(1-\rho)(7+2\rho)^2 - (7-2\rho)^2 = \rho(7 - 28\rho - 4\rho^2) \ge 0, \qquad k(1/5)^2 < 2/7.$$

In every case considered here, $AB > 5$. Set

$$k_1 = \frac{1}{2} + \frac{1}{7AB}, \quad k_2 = \frac{1}{2} + \frac{1}{7B^2}, \quad K_1 = k_1^2 k_2, \quad L_1 = 6K_1 - \frac{1}{2}, \quad Q_1 = 9k_1^2 k_2^2.$$

Every cubic monomial has at most two occurrences of edges incident to the smallest entry, counting repeated edges. Expansion about the three leading factors in (2.9) therefore bounds its remainder by $(K_1 - 1/8)\prod |z_{ij}|$. The sums of the absolute cubic coefficients are 2 for terms contributing to $a_i/\prod_{j\neq i} a_j$, and 6 for those contributing to $1/(a_i a_j)$. Combining these with (2.8), and observing that each quartic monomial has exactly two incident edge occurrences, gives

$$1 \le |\det M| \le 2K_1\left(\frac{T}{ABS} + \frac{S}{ABT} + \frac{B}{AST} + \frac{A}{BST}\right) + L_1 \sum_{i<j} \frac{1}{A_i A_j} + \frac{Q_1}{ABST}. \tag{2.10}$$

Put $\tau = T/(ABS)$. For fixed $\tau > 0$, substituting $T = \tau ABS$ and $S \ge B$ gives

$$\frac{S}{ABT} + \frac{B}{AST} + \frac{A}{BST} \le \frac{1}{\tau}\left(\frac{2}{A^2B^2} + \frac{1}{B^4}\right),$$

$$\sum_{i<j} \frac{1}{A_i A_j} \le \frac{2}{AB} + \frac{1}{B^2} + \frac{1}{\tau}\left(\frac{1}{A^2B^2} + \frac{2}{AB^3}\right), \qquad \frac{1}{ABST} \le \frac{1}{\tau A^2 B^4}.$$

In these positive expressions, replace A, B by rational lower bounds α, B_0. Evaluate k_1, k_2, and hence K_1, L_1, Q_1, at $A = \alpha, B = B_0$. Let

$$P_0 = \frac{2}{\alpha B_0} + \frac{1}{B_0^2}, \qquad J_0 = 2K_1\left(\frac{2}{\alpha^2 B_0^2} + \frac{1}{B_0^4}\right) + L_1\left(\frac{1}{\alpha^2 B_0^2} + \frac{2}{\alpha B_0^3}\right) + \frac{Q_1}{\alpha^2 B_0^4}.$$

Equation (2.10) now bounds the determinant by the convex function

$$F(\tau) = 2K_1\tau + L_1 P_0 + \frac{J_0}{\tau}.$$

If $A \ge 3$, take $\alpha = B_0 = 3$. Lemma 2.3 gives $\tau > 2$, while direct rational substitution gives $F(2) < 7/10$ and $F(3) < 19/20$. Convexity excludes $2 < \tau \le 3$.

If $A < 3$, use the lower bounds in the table. Lemma 2.2 gives $\tau > 2/A > 2/3$. For all five explicitly listed triples (α, B_0, t), direct rational substitution gives

$$F(2/3) < \frac{1}{2}, \qquad F(t) < \frac{499}{500} < 1.$$

The largest of the five right endpoint values is $964921210337222547/967173115740160000 < 499/500$, in the norm-two row. The rational lower bounds for A follow by squaring. All satisfy $\alpha B_0 > 5$. Convexity excludes $2/3 < \tau \le t$, proving all rows at once. □

3 Interpolation of square roots

We construct integral combinations of square roots by interpolation. The interpolation weights cancel the low-order terms involving the fixed entries, leaving an explicit remainder. When these combinations are nonzero, bounds for them give gaps between the last two entries. We also bound the ranges in which these combinations can vanish.

Let $a_1, \dots, a_m, p, q$ be distinct tuple entries, $m = 2, 3, 4$, with $1 \le A_1 \le \cdots \le A_m \le P = |p| \le Q = |q|$ and $A_1 P > 1$. Choose $\eta^2 = pq$, put $x = 1/p, y = 1/q$, and use $f(w) = \sqrt{1+w}$ on its analytic branch at zero. Define

$$z = \eta f(xy), \qquad R_i = a_i \eta f(x/a_i) f(y/a_i), \qquad \Delta_m = \prod_{i<j} (a_i - a_j).$$

Their squares are $pq+1$ and $(a_i p + 1)(a_i q + 1)$. Thus $z \in \mathcal{O}_K$. Each R_i equals, up to sign, the product of integral square roots of $a_i p + 1$ and $a_i q + 1$, so $R_i \in \mathcal{O}_K$ as well. We call the fixed entries $a_1, \dots, a_m$ the anchors. With the common branch choices above, define

$$L_m = -\Delta_m z + \sum_{i=1}^{m} \frac{\Delta_m a_i^{m-2}}{\prod_{j \ne i}(a_i - a_j)} R_i \ \in \mathcal{O}_K.$$

The coefficients are integer polynomials, since each denominator divides the Vandermonde product up to sign. A nonzero L_m has modulus at least one.

Let h_j be the complete homogeneous symmetric polynomial of degree j in $1/a_1, \dots, 1/a_m$, and write $f(xW) f(yW) = \sum_{n \ge 0} f_n(x, y) W^n$. Partial fractions of $\prod_i (1 - W/a_i)^{-1}$ give

$$\sum_i \frac{a_i^{m-1-n}}{\prod_{j \ne i}(a_i - a_j)} = \begin{cases} 1 & n = 0, \\ 0 & 1 \le n < m, \\ (-1)^{m-1} h_{n-m} / \prod_i a_i & n \ge m. \end{cases}$$

Hence the exact identity is

$$\frac{L_m}{\Delta_m \eta} = 1 - f(xy) + \frac{(-1)^{m-1}}{\prod_i a_i} F_m, \qquad F_m = \sum_{n \ge m} f_n(x, y) h_{n-m}. \tag{3.1}$$

All series converge absolutely. Indeed, for $b_n = \binom{1/2}{n}$,

$$\left(\sum_{n \ge 0} |b_n| W^n \right)^2 = (2 - \sqrt{1 - W})^2 = 5 - W - 4\sqrt{1 - W},$$

so $|f_n(x, y)| \le 4 |b_n| P^{-n}$ for $n \ge 2$. As h_j has $\binom{j+m-1}{m-1}$ monomials, and $|b_n|$ decreases for $n \ge 1$, we obtain the complete remainder estimate

$$|F_m - f_m| \le 4 \left| \binom{1/2}{m+1} \right| P^{-m} \left[\left(1 - \frac{1}{A_1 P} \right)^{-m} - 1 \right]. \tag{3.2}$$

Direct multiplication gives

$$f_2 = -\frac{(x-y)^2}{8}, \qquad f_3 = \frac{(x^2 - y^2)(x - y)}{16}, \qquad f_4 = -\frac{(x-y)^2 (5x^2 + 6xy + 5y^2)}{128}. \tag{3.3}$$

Whenever $PQ > 10000$, rationalization and $\operatorname{Re} \sqrt{1+w} \ge \sqrt{1 - |w|}$ give

$$\frac{99}{100} < |f(xy)| < \frac{101}{100}, \qquad \frac{100}{201 PQ} < |1 - f(xy)| < \frac{100}{199 PQ}. \tag{3.4}$$

Lemma 3.1. *Let (H, λ, U) be either row of the table*

H	λ	U
32	1/10	179/10
48	19/200	15

Suppose $A_1 P \ge H,\ Q \ge 4P,\ PQ > 10000$. For two anchors of moduli $X \le Y$, and $Z = |z|$,

$$|L_2| < \frac{21Y}{20Z} + \frac{Z}{2XP^2}.$$

For three anchors of moduli X, Y, V,

$$|L_3| < \frac{51|\Delta_3|}{100Z} + \lambda\frac{|\Delta_3|Z}{XYVP^3}, \qquad L_3 = 0 \implies 5XYVP^2 < Q < UXYVP^2.$$

Proof. For two anchors, (3.3) and the tail estimate bound $P^2|F_2|$ by

$$\frac{25}{128} + \frac{1}{4}\left[\left(\frac{H}{H-1}\right)^2 - 1\right].$$

Equation (3.1), $|\Delta_2| \le 2Y$, and (3.4) give the required estimate because

$$\frac{202}{199} < \frac{21}{20}, \qquad \frac{200}{99}\left\{\frac{25}{128} + \frac{1}{4}\left[\left(\frac{H}{H-1}\right)^2 - 1\right]\right\} < \frac{1}{2}.$$

For three anchors, $|y/x| \le 1/4$ gives $45/1024 \le P^3|f_3| \le 85/1024$. Put

$$r_H = \frac{5}{32}\left[\left(\frac{H}{H-1}\right)^3 - 1\right], \quad l_H = \frac{45}{1024} - r_H, \quad u_H = \frac{85}{1024} + r_H.$$

Then $l_H/P^3 \le |F_3| \le u_H/P^3$, with $l_H > 0$. The required rational comparisons are

$$\frac{101}{199} < \frac{51}{100}, \quad \frac{100u_H}{99} < \lambda, \quad \frac{100}{201u_H} > 5, \quad \frac{100}{199l_H} < U.$$

For clarity, the last two exact fractions are

H	$100/(201u_H)$	$100/(199l_H)$
32	610119680/120943911	122023936/6879629
48	2126295040/398301399	2126295040/142842001

Substitution in the exact identity proves both the direct estimate and, when $L_3 = 0$, the two strict bounds for Q. □

Lemma 3.2. *In the two-anchor situation of Lemma 3.1, assume also $Y \ge 3,\ P \ge 5Y$. If $Q \ge 5XYP$, then*

$$Q > \frac{X^2P^3}{h_0}, \qquad h_0 = \frac{10001}{10000}.$$

Proof. Here $L_2 = R_a - R_b - (a-b)z$. If it vanishes, subtracting the identities for R_a^2, R_b^2 and then squaring gives

$$(a+b-p-q)^2 = 4(ab+1)(pq+1).$$

Solving this quadratic in q gives

$$q = a + b + p + 2abp \pm 2\sqrt{(ab+1)(ap+1)(bp+1)}.$$

Since $XY \geq 3, XP \geq 15, YP \geq 45$,

$$\frac{Q}{XYP} \leq \frac{19}{45} + 2 + 2\sqrt{\frac{4}{3}\frac{16}{15}\frac{46}{45}} < 5.$$

The last comparison is $11776 < 13456$ after squaring. Thus $Q \geq 5XYP$ makes $L_2 \neq 0$. Moreover

$$\frac{Z^2}{Y^2} \geq \frac{5XP^2}{Y} - \frac{1}{Y^2} \geq 5XP - 1 \geq 159.$$

The first term in the bound of Lemma 3.1 is less than $1/2$. Since $1 \leq |L_2|$, the second term exceeds $1/2$, giving $Z > XP^2$. Finally $Z^2 \leq PQ + 1 < h_0 PQ$, proving the assertion. □

Lemma 3.3. *For three anchors $A \leq B \leq S$, suppose $P \geq 4S,\ AP \geq 9,\ Q \geq P$. Then*

$$|L_3| < \frac{5BS^2}{Z} + \frac{3SZ}{AP^3}.$$

Proof. The absolute coefficient series gives $|f_n| \leq P^{-n}/4$ for $n \geq 3$, and hence

$$|F_3| \leq \frac{1}{4P^3(1 - 1/(AP))^3}.$$

Also $P^2 \geq 4AP \geq 36$, $|\Delta_3| \leq 8BS^2$, $35/36 < |f(xy)| < 37/36$, and $|1 - f(xy)| < 36/(71PQ)$. The two contributions in (3.1) are bounded by $(296/71)BS^2/Z$ and $(6561/2240)SZ/(AP^3)$, respectively. □

Lemma 3.4. *For three anchors $A \leq B \leq S \leq P$, if $P \geq 1000,\ Q \geq 4P$, then*

$$|L_3| < \frac{5BS^2}{Z} + \frac{3SZ}{4AP^3}, \qquad L_3 = 0 \implies 5ABSP^2 < Q < 15ABSP^2.$$

Proof. Use the calculation in the proof of Lemma 3.1 with $r_{1000}, l_{1000}, u_{1000}$. Since $|\Delta_3| \leq 8BS^2$, the needed direct comparisons are $808/199 < 5$ and $800u_{1000}/99 < 3/4$. For $L_3 = 0$, the required rational comparisons are $100/(201u_{1000}) > 5$ and $100/(199l_{1000}) < 15$. □

Lemma 3.5. *For ordered anchors $1 \leq A \leq B \leq S \leq T \leq P \leq Q$, assume $AP > 1$ and $P^2 \geq 32BST/A^3$. The coherent forms L_3 on (a, b, s) and (a, b, t), using the same z, R_a, R_b, cannot both vanish.*

Proof. If both vanished, the quadratic polynomials interpolating uR_u at the two triples would have leading coefficient z and agree at a, b, so they would coincide. Write the common polynomial $J(X) = zX^2 + vX + w$. Since $z^2 - pq = 1$,

$$J(X)^2 - X^2(pX + 1)(qX + 1) = (X - a)(X - b)(X - s)(X - t).$$

For the elementary symmetric functions e_j of the anchors, $w^2 = e_4,\ 2vw = -e_3,\ v^2 + 2zw - 1 = e_2$. Thus

$$pq + 1 = \frac{(e_2 + 1 - e_3^2/(4e_4))^2}{4e_4}.$$

Using $|e_2| \leq 6ST,\ |e_3| \leq 4BST,\ |e_4| = ABST$, this gives

$$|pq + 1| \leq \frac{(6ST + 1 + 4BST/A)^2}{4ABST} \leq \frac{121BST}{4A^3}.$$

But with $r = BST/A^3 \geq 1$, $|pq + 1| \geq PQ - 1 \geq P^2 - 1 \geq 32r - 1 > 121r/4$, a contradiction. □

Lemma 3.6. *For four anchors $A \le B \le S \le T \le P$, if $P \ge 1000$, $Q \ge 1000P$, then*

$$|L_4| < \frac{9BS^2T^3}{Z} + \frac{4ST^2Z}{5AP^4}, \qquad L_4 = 0 \implies 10ABSTP^3 < Q < 17ABSTP^3.$$

Proof. For nonzero complex numbers $u_1, \dots, u_m$, ordered by modulus,

$$\left| \prod_{i<j} (u_i - u_j) \right| \le m^{m/2} \prod_j |u_j|^{j-1}.$$

Indeed, after division by $\prod_j u_j^{j-1}$, the expression is, up to sign, the polynomial $\prod_{i<j}(1 - z_i \cdots z_{j-1})$, $z_i = u_i/u_{i+1}$. Applying the maximum-modulus principle to each variable reduces the problem to $|z_i| = 1$. The numbers u_j can then be scaled to have modulus one. Hadamard's inequality for their Vandermonde matrix proves the bound. Thus $|\Delta_4| \le 16BS^2T^3$.

Put $k = 1/1000$ and

$$r = \frac{7}{64}\left[\left(\frac{1000}{999}\right)^4 - 1\right], \quad l = \frac{(1-k)^2(5-6k-5k^2)}{128} - r, \quad u = \frac{(1+k)^2(5+6k+5k^2)}{128} + r.$$

Formula (3.3) and (3.2) give $l/P^4 \le |F_4| \le u/P^4$, with $l > 0$. The exact identity proves the assertions using

$$\frac{16 \cdot 101}{199} < 9, \quad \frac{1600u}{99} < \frac{4}{5}, \quad \frac{100}{201u} > 10, \quad \frac{100}{199l} < 17.$$

All four are strict rational comparisons. □

3.1 A gap estimate for the last two entries

Lemma 3.7. *Let six ordered entries have $1 \le A \le B \le S \le T \le C \le D$, $B \ge 3$, $T > 2ABS$. Choose either row (H, λ, U) of Lemma 3.1, and suppose $AT \ge H$. Put*

$$\mathcal{Q} = \frac{|(a-b)(a-s)(b-s)|}{BS^2}, \quad \mu = \frac{51051}{400000}, \quad c_\lambda = \frac{9999}{10000\lambda^2}.$$

Then

$$C > c_\lambda \left(\frac{1}{\mathcal{Q}} - \frac{\mu}{A^2B^2}\right)^2 \frac{A^2T^5}{S^2} > \frac{A^2T^5}{S^2}, \tag{3.5}$$

$$D > \frac{A^2C^5}{S^2}, \qquad D > \frac{A^2C^7}{S^2T^4}. \tag{3.6}$$

Moreover $C > 1000$, $C > 200S$, $C > 4T$, $AC > 1000$, $D > 1000C$. The conclusions for C use only the first five entries.

Proof. Lemma 2.4 gives $C > 3ABST$. Since $T > 18$ and $AT \ge H \ge 32$,

$$C > 27H \ge 864, \quad TC > 486H > 10000, \quad C/T > 27, \quad C/S > 9H > 200.$$

Also $T > 5B, T > 5S$, and $C > 5ABT$. Lemma 3.2, applied to a, b, t, c, gives $C > A^2T^3/h_0 > 18^3/h_0 > 1000$. As

$$\frac{A^2T^2}{BS} > 4A^4BS \ge 36 > 5h_0,$$

we have $C > 5BST$, so a second application on b, s, t, c gives $C > B^2T^3/h_0$. The form L_3 on a, b, s, evaluated at t, c, cannot vanish, since

$$\frac{BT}{AS} > 2B^2 \geq 18 > Uh_0$$

for both parameter rows, contradicting the upper bound for C when $L_3 = 0$.

Let $W = |\sqrt{tc+1}|$ and $k = 3999/4000$. From the second application of Lemma 3.2 and $W^2 > kTC$,

$$X_0 := \frac{5BS^2}{W} < 5\sqrt{h_0/k}\frac{S^2}{T^2} < \frac{5}{4A^2B^2}\sqrt{h_0/k} < \frac{5005}{4000A^2B^2}.$$

The first term in the L_3 estimate of Lemma 3.1 is $(51/500)\mathcal{Q}X_0 < \mu\mathcal{Q}/(A^2B^2)$. Since L_3 is a nonzero algebraic integer, the preceding estimate gives

$$W > \alpha\frac{AT^3}{S}, \qquad \alpha = \lambda^{-1}\left(\frac{1}{\mathcal{Q}} - \frac{\mu}{A^2B^2}\right).$$

Since $0 < \mathcal{Q} \leq 8$, $AB \geq 3$, and $\lambda \leq 1/10$, $\alpha > 1$ and $(9999/10000)\alpha^2 > 1$. Also $\alpha^2A^2T^6/S^2 > T^4 > 18^4 > 10000$. After squaring, subtracting one causes a relative loss of less than $1/10000$, proving (3.5).

Repeat this five-entry argument on a, b, s, c, d. Here $C > 2ABS$ and $AC \geq H$. This gives the first inequality in (3.6) and $D/C > A^2C^4/S^2 \geq C^2 > 1000$. The form L_4 on (a, b, s, t), evaluated at (c, d), satisfies the hypotheses of Lemma 3.6. If it vanished, that lemma would give $D < 17ABSTC^3$. Together with $D > A^2C^5/S^2$, this would imply $C^2 < 17BS^3T/A$, contrary to

$$\frac{C^2}{17BS^3T/A} > \frac{9}{17}A^3\frac{BT}{S} > \frac{18}{17}A^3B^2 > 1.$$

For its first summand, the preceding lower bound for D and Lemma 2.4 give

$$\left(\frac{9BS^2T^3}{|\sqrt{cd+1}|}\right)^2 < \frac{1}{9kA^8B^4} < \frac{1}{100}.$$

Thus $|\sqrt{cd+1}| > (9/8)AC^4/(ST^2)$. Squaring and subtracting one proves the last inequality in (3.6), because $(17/64)A^2C^8/(S^2T^4) > 1$ follows already from $C > 3ABST$. The preceding estimates also give all the stated size inequalities. □

4 Simultaneous approximation

We construct simultaneous approximations using two or three fixed tuple entries. All polynomial divisions below are performed over $\mathbb{Z}$ before specializing the fixed entries. No unique factorization assumption on $\mathcal{O}_K$ is needed.

4.1 A common residue construction

Lemma 4.1. *Let $a_1, \ldots, a_m, c, d$, where $m \geq 2$, be distinct nonzero members of a $D(1)$-tuple in $\mathcal{O}_K$. Put*

$$A_j = |a_j|, \quad A = \min_j A_j, \quad S = \max_j A_j, \quad C = |c| > 1, \quad AC > 1,$$

$$H=\prod_{j=1}^{m}a_j,\qquad \Delta=\prod_{1\le i<j\le m}(a_i-a_j),\qquad \eta=1-\frac{1}{AC},\qquad e(X)=\sqrt{1+\frac{1}{XC}}\left(1+\frac{X}{C}\right).$$

For every integer $k\ge 1$ there is an explicit matrix $Q_k=(Q_{ijk})_{0\le i,j\le m}$ of integral polynomials in $a_1,\dots,a_m$ and c. Its jth column, for $j\ge 1$, is divisible by $a_j^{(m-1)k}$, and

$$\det Q_k=\left((-1)^{m(m+1)/2}4^{m+1}H^{m-1}\Delta^{2m}\right)^k\prod_{v=-1}^{m-1}\binom{1/2-vk}{k}\neq 0. \tag{4.1}$$

With $\theta_0=1$, $\theta_j=\sqrt{1+1/(a_jc)}$ defined by the binomial series,

$$\left|\sum_{j=0}^{m}Q_{ijk}\theta_j\right|\le V_{m,k}:=\left(\frac{4|\Delta|^2}{|H|C^m}\right)^k\eta^{-mk+1/2}\frac{\Gamma(k+3/2)\Gamma(mk-1/2)}{\pi\Gamma((m+1)k+1)}. \tag{4.2}$$

If $z\in\mathcal{O}_K$, $z^2=cd+1$, and $Z=|z|$, then

$$1\le\frac{Ce(S)}{Z}\max_i\sum_{j=1}^{m}\frac{|Q_{ijk}|}{A_j}+V_{m,k}Z. \tag{4.3}$$

At $k=1$ the left side can be replaced by 2.

Proof. Set $\alpha_0=0$, $\alpha_j=H/a_j$, and $N=Hc$. With δ_{ij} the Kronecker delta, define

$$f_{ij}=\sum_{\substack{h_0,\dots,h_m\ge 0\\ h_0+\cdots+h_m=k+\delta_{ij}-1}}\binom{k+1/2}{h_j}N^{-k}(N+\alpha_j)^{k-h_j}\prod_{\ell\neq j}\binom{-k-\delta_{i\ell}}{h_\ell}(\alpha_j-\alpha_\ell)^{-k-\delta_{i\ell}-h_\ell}, \tag{4.4}$$

and

$$Q_{ijk}=(4H^mc\Delta^2)^kf_{ij}.$$

The residue theorem, applied at the interpolation points, gives

$$\frac{1}{2\pi\mathrm{i}}\int_\Gamma\frac{(1+w/N)^{k+1/2}}{(w-\alpha_i)\prod_{\ell=0}^{m}(w-\alpha_\ell)^k}\,dw=\sum_{j=0}^{m}f_{ij}\theta_j. \tag{4.5}$$

Here Γ surrounds the interpolation points and avoids the ray from $-N$ away from 0. Such a contour exists because $|\alpha_j/N|\le 1/(AC)<1$. Formula (4.4) follows by differentiating the numerator and the other pole factors at each pole. This residue construction appears in [9, §7].

We prove integrality term by term. Apart from a sign and the scalar

$$4^k\binom{k+1/2}{h_j}\prod_{\ell\neq j}\binom{-k-\delta_{i\ell}}{h_\ell},$$

a summand in column zero is

$$c^{k-h_0}\Delta^{2k}\prod_{r=1}^{m}a_r^{\delta_{ir}+h_r},$$

and a summand in column $j\ge 1$ is

$$a_j^{(m-1)k}(a_jc+1)^{k-h_j}\prod_{\substack{\ell\neq j\\ \ell\ge 1}}a_\ell^{\delta_{i\ell}+h_\ell}\frac{\Delta^{2k}}{\prod_{\substack{\ell\neq j\\ \ell\ge 1}}(a_\ell-a_j)^{k+\delta_{i\ell}+h_\ell}}. \tag{4.6}$$

For $\ell \neq j$ one has $\delta_{i\ell} + h_\ell \leq k$, and $h_j \leq k$. Thus all factors $a_\ell - a_j$ in the denominators cancel, and every remaining exponent is nonnegative. Also $4^{h_j}\binom{k+1/2}{h_j} \in \mathbb{Z}$. To see this, expand $(1+x)^k(1+x)^{1/2}$ and use $\binom{1/2}{r} = (-1)^{r-1}2C_{r-1}/4^r$ for $r \geq 1$, where C_r is a Catalan number. This proves polynomial integrality and the claimed column divisibility. At $k = 1$ the scalar factors are 4, 6, or multiples of 4, so every entry is even.

The determinant identity for the residue coefficients is

$$\det(f_{ij}) = N^{-(m+1)k}\left(\prod_{v=-1}^{m-1}\binom{1/2 - vk}{k}\right)\prod_{\ell=0}^{m}\prod_{\substack{r=0\\ r\neq\ell}}^{m}(\alpha_r - \alpha_\ell)^{-k}.$$

This is the explicit identity in [2, p. 1721], with $\nu = 1/2$ and $x = 1/N$. Although stated there for distinct integer interpolation points and real x, the identity extends to complex points. Substitute the rational expressions (4.4) and clear all denominators. The result is a polynomial identity, valid for all distinct integer points and an interval of real x, and hence identically valid over $\mathbb{C}$. The product of the pairwise differences of the interpolation points is, up to sign, $H^{m(m-1)/2}\Delta$. Multiplication by the clearing factor gives (4.1). All its factors are nonzero. Each numerator factor in the displayed binomial coefficients is a nonzero half-integer.

Deform the contour in (4.5) onto $w = -N(1+x)$, $x \geq 0$. The numerator's branch jump has modulus $2x^{k+1/2}$, while every pole factor after division by N has modulus at least $x + \eta$. The large-circle contribution vanishes because the integrand is $O(|w|^{-mk-1/2})$, and the small-circle contribution at the branch point also tends to zero. Multiplying by the clearing factor therefore bounds the residual by

$$\left(\frac{4|\Delta|^2}{|H|C^m}\right)^k \frac{1}{\pi}\int_0^\infty \frac{x^{k+1/2}}{(x+\eta)^{(m+1)k+1}}\,dx.$$

The substitution $x = \eta y$ and the beta integral give (4.2).

Choose integral roots $r_j^2 = a_j c + 1$, $v_j^2 = a_j d + 1$ and put $u_j = \pm r_j v_j/(a_j z)$. Here $z \neq 0$, since otherwise $|d| = 1/C < 1$. Choose each sign so that $|\theta_j + u_j| \geq |\theta_j|$. Factoring

$$\theta_j^2 - u_j^2 = \frac{(a_j c + 1)(a_j - c)}{a_j^2 c z^2}$$

gives

$$|\theta_j - u_j| \leq \frac{Ce(A_j)}{A_j Z^2} \leq \frac{Ce(S)}{A_j Z^2}.$$

The last inequality follows from $(e(X)^2)' = [2 + 2X/C - 1/X^2 + 1/C^2]/C > 0$ for $X \geq 1$. The vector with coordinates

$$q = Hz, \qquad n_j = (H/a_j)(\pm r_j v_j)$$

is nonzero. Column divisibility makes every row form $Q_{i0k}q + \sum_{j\geq 1} Q_{ijk}n_j$ a multiple of H in $\mathcal{O}_K$. Nonsingularity makes some row form nonzero, so its modulus is at least $|H|$. Division by $|H|$ and the triangle inequality prove (4.3). At $k = 1$ all these forms are multiples of $2H$, proving the stronger assertion. □

4.2 Two fixed entries at every order

Proposition 4.2. *Let a, b, c, d be distinct nonzero members of a $D(1)$-tuple, with $1 \le A = |a| \le B = |b|$, $C = |c| > 4B$ and $AC > 1000$. Put $\delta = |a - b|$ and*

$$H_2 = 42B^3C, \qquad R_2 = \frac{27AB(AC-1)^2}{16A^2\delta^2} > \frac{2AC^2}{5B}.$$

For $z^2 = cd + 1$, $Z = |z|$, and every integer $k \ge 1$,

$$1 < \frac{3CH_2^k}{BZ} + \frac{Z}{2R_2^k}. \tag{4.7}$$

If $R_2 > 1$ and $\lambda = 1 + \log H_2/\log R_2 < 2$, then

$$Z^{2-\lambda} < 253B^2C^2. \tag{4.8}$$

Proof. Use Lemma 4.1 with $m = 2$. Its interpolation points are $0, b, a$ and its integral matrix has

$$\det Q_k = (-64ab(a-b)^4)^k \prod_{v=-1}^{1} \binom{1/2 - vk}{k} \neq 0.$$

We bound columns 1 and 2 using local residue integrals. At either nonzero interpolation point let $u \le v$ be its two distances to the other points, and integrate on the circle of radius $2u/5$. The product of the three pole distances is at least

$$\frac{6u^2(v - 2u/5)}{25}.$$

The additional factor $|w - \alpha_i|$ is at least $2u/5$, which cancels the factor given by the circle length divided by 2π. Also $|w| \le 7B/5$. Put

$$F_2 = 1 + \frac{7}{5AC}, \qquad p_2 = \sqrt{\frac{F_2}{1 - 1/(AC)}}.$$

The modulus of the numerator, divided by $|\theta_j|$, is at most $p_2F_2^k$. The circles lie in $|w| < |abc|$, so they support the specified analytic branch.

At the point a the two distances are A, δ, and at b they are B, δ. In both cases

$$\frac{v^2}{v - 2u/5} \le \frac{5B}{2}. \tag{4.9}$$

Indeed, if $\delta \le A$ at a, or $\delta \le B$ at b, the ratio is at most $5B/3$. Otherwise set $x = A/B \le 1$ and use $\delta \le A + B$. The two ratios divided by B are bounded by $(1 + x)^2/(1 + 3x/5)$ and $(1 + x)^2/(x + 3/5)$, respectively. Both functions increase on $[0, 1]$ and equal $5/2$ at 1.

For column 2, whose pole is a, the local estimate before (4.9) is

$$|Q_{i2k}| \le p_2 \left(\frac{50}{3} B^2 C F_2 \frac{v^2}{v - 2u/5}\right)^k.$$

Column 1 has the same expression with A^2 in place of B^2 and the distances taken at b. These formulas follow by multiplying the integral estimate by $(4a^2b^2c(a-b)^2)^k$ and using $uv = A\delta$ at a and $uv = B\delta$ at b. Since $(125/3)(1 + 7/5000) < 42$, they imply

$$|Q_{i2k}| < p_2H_2^k, \qquad |Q_{i1k}| < p_2\Big((A/B)^2H_2\Big)^k.$$

Consequently

$$\frac{Ce(B)}{Z}\sum_{j=1}^{2}\frac{|Q_{ijk}|}{A_j} < \frac{2p_2 e(B)CH_2^k}{BZ} < \frac{3CH_2^k}{BZ},$$

where $A_1 = A, A_2 = B$. The last strict comparison follows, for example, by squaring and using

$$p_2^2 \le \frac{5007}{4995}, \qquad e(B)^2 < \frac{25\cdot 1001}{16000}, \qquad 4\frac{5007}{4995}\frac{25\cdot 1001}{16000} < 9.$$

We next bound the residual uniformly for all $k \ge 1$. Write $\eta = 1 - 1/(AC)$. The exact beta bound in (4.2) gives

$$V_{2,k}R_2^k = \eta^{1/2}b_k, \qquad b_k = \left(\frac{27}{4}\right)^k \frac{\Gamma(k+3/2)\Gamma(2k-1/2)}{\pi\Gamma(3k+1)}.$$

Here $b_1 = 27/64$, and

$$\frac{b_{k+1}}{b_k} = \frac{(27/4)(k+3/2)(2k-1/2)(2k+1/2)}{(3k+1)(3k+2)(3k+3)} < 1,$$

because denominator minus numerator is $(432k^2 + 1110k + 273)/32 > 0$. Thus $V_{2,k} < (27/64)R_2^{-k} < (1/2)R_2^{-k}$ for every $k \ge 1$. The root-transfer inequality (4.3) now gives (4.7). Finally $\delta \le 2B$ and $(27/64)(999/1000)^2 > 2/5$ give the stated lower bound for R_2.

For (4.8), choose $k = 1 + \lfloor \log Z / \log R_2 \rfloor$. Then $Z/(2R_2^k) < 1/2$ and $H_2^k \le H_2 Z^{\lambda-1}$, whence

$$Z^{2-\lambda} < 6(C/B)H_2 = 252B^2C^2 < 253B^2C^2.$$

□

Corollary 4.3. *Under the preceding hypotheses, suppose $D = |d| \ge C$ and put $x = \log_{10} C$. Let rational h, r, u satisfy*

$$\log_{10} H_2 < x + h, \qquad \log_{10} R_2 > 2x + r, \qquad \log_{10}(253B^2) < u.$$

Set $t = h - r$, $\beta = 4u + 8h - 4r$ and $\gamma = 2ur + \beta t$. If $x > t$ and $2x + r > 0$, then

$$\log_{10} D < 7x + \beta + \frac{\gamma}{x - t} + \frac{2}{5}.$$

Proof. The assumptions give $R_2 > 1$ and $2 - \lambda > (x - t)/(2x + r) > 0$. Since $CD > 2$, $CD < 2Z^2$. Apply (4.8) and $\log_{10} 2 < 2/5$ to obtain

$$\log_{10} D < \frac{2}{5} - x + \frac{2(2x+u)(2x+r)}{x-t} = 7x + \beta + \frac{\gamma}{x-t} + \frac{2}{5}.$$

□

4.3 Three fixed entries at order two

Lemma 4.4. *Let a, b, s, c, d be distinct nonzero members of a $D(1)$-tuple, with*

$$1 \le A = |a| \le B = |b| \le S = |s|, \qquad C = |c| \ge 4S, \qquad AC \ge 1000.$$

For $z^2 = cd + 1$ and $Z = |z|$ one has

$$1 < \frac{8}{9}\,608^2\frac{B^3S^8C^3}{Z} + \frac{3920}{81}\frac{B^2S^5Z}{A^3C^6}. \tag{4.10}$$

If $D = |d| \geq C$, the closed interval

$$\frac{512}{81}608^4 B^6 S^{16} C^5 \leq D \leq \frac{A^6 C^{11}}{2 \cdot 28^4 (10/81)^2 B^4 S^{10}} \tag{4.11}$$

is impossible.

Proof. Use the common construction with $m = 3, k = 2$, write $H = abs$, $\Delta = (a-b)(a-s)(b-s)$, and divide rows by

$$d_0 = \Delta, \quad d_1 = a(b-s), \quad d_2 = b(a-s), \quad d_3 = s(a-b).$$

Let $R_{ij} = Q_{ij2}/d_i$. These are integral polynomials. To check this directly in (4.6), in the row indexed by 0 every incident difference retains an exponent at least one. In row $i \geq 1$ and column $j \neq i, 0$, the factor a_i is explicit, and the opposite difference retains an exponent at least one because $\delta_{i\ell} = 0$ and $h_\ell \leq 1$ at its other endpoint. In column i, the factor a_i^4 supplies a_i, and the opposite difference has exponent 4. The entries in the column indexed by 0 also contain all the required row factors. For $j \geq 1$, every entry in column j of R remains divisible by a_j, and

$$\det R = -226800 H^3 \Delta^{10} \neq 0,$$

by (4.1) and $\prod_{i=0}^{3} d_i = H\Delta^2$.

We now bound the coefficients after these row divisions. Put

$$\rho_j = \min\left(1, \min_{\substack{\ell \neq j \\ \ell \geq 1}} \frac{|a_\ell - a_j|}{A_\ell}\right), \qquad \sigma_j = \frac{|H|\rho_j}{A_j},$$

$$F = 1 + \frac{5}{4AC}, \quad p = \sqrt{\frac{F}{1 - 1/(AC)}}, \quad K_0 = \frac{1024}{27}, \quad G = 16 K_0 F.$$

At the pole α_j integrate on the circle of radius $\sigma_j/4$. It lies in the disk $|w| < |N|$ and misses every other pole. The product of its four pole distances is at least

$$\frac{1}{4}\left(\frac{3}{4}\right)^3 \sigma_j \prod_{\ell \neq j} |\alpha_j - \alpha_\ell|.$$

Retaining the factor $|w - \alpha_i|$ in the denominator gives

$$|Q_{ij2}| \leq p \tau_{ij} M_j^2, \qquad M_j = K_0 F C |\Delta|^2 \frac{A_j^3}{\rho_j \prod_{\ell \neq j,\, \ell \geq 1} |a_\ell - a_j|},$$

where

$$\tau_{jj} = 1, \qquad \tau_{0j} \leq \rho_j/3, \qquad \tau_{ij} \leq \frac{A_i \rho_j}{3|a_i - a_j|} \quad (i \neq j,\ i, j \geq 1).$$

For $i \neq j$ and w on this circle, the ratio $(\sigma_j/4)/|w - \alpha_i|$ is at most $\sigma_j/(3|\alpha_i - \alpha_j|)$. The numerator divided by $|\theta_j|$ is at most pF^2, which accounts for all remaining factors.

The ordered bounds $|a - b| \leq 2B$ and $|a - s|, |b - s| \leq 2S$ imply

$$M_j \leq 16 K_0 F C B^2 S^5 = G C B^2 S^5. \tag{4.12}$$

For the term 1 in $1/\rho_j$, the remaining product is $A_j^3|a_j - a_\ell||a_j - a_r||a_\ell - a_r|^2$, at most $16B^2S^5$. For either other term in $1/\rho_j$, one incident difference cancels and the bound is $8B^2S^5$.

Use $M_j^2 \le (GCB^2S^5)M_j$ in the weighted row sum. For row $i \ge 1$ its diagonal contribution is

$$\frac{M_i}{A_i|d_i|} = K_0FC\frac{A_i|\Delta|}{\rho_i} \le 8K_0FCBS^3.$$

The three alternatives in $1/\rho_i$ give bounds $8BS^3, 4BS^3, 4BS^3$ for the remaining product. With r the remaining index, each off-diagonal contribution satisfies

$$\frac{\tau_{ij}M_j}{A_j|d_i|} \le \frac{K_0FC}{3}A_j^2|a_i - a_r|^2 \le \frac{4}{3}K_0FCBS^3.$$

For the row indexed by 0 the sum is at most

$$\frac{K_0FC}{3}\sum_j A_j^2|a_\ell - a_r| \le 2K_0FCBS^2.$$

Therefore every row has weighted sum at most $(32/3)K_0FCBS^3$. Multiplying by the common bound GCB^2S^5 from (4.12) gives

$$\sum_{j=1}^{3}\frac{|R_{ij}|}{A_j} \le \frac{2}{3}pG^2C^2B^3S^8. \tag{4.13}$$

For the residual, the common beta integral at $m = 3, k = 2$ can be written

$$\left|\sum_{j=0}^{3} Q_{ij2}\theta_j\right| \le \frac{10}{81}\eta^{1/2}\left(\frac{27|\Delta|^2}{64|H|C^3\eta^3}\right)^2, \qquad \eta = 1 - 1/(AC).$$

The half-integer gamma identities give the factor $10/81$ directly. For $i = 0, 1, 2, 3$ the respective bounds for $|\Delta|^2/(|Hd_i|)$ are

$$\frac{8S}{A}, \quad \frac{32BS^2}{A^2}, \quad \frac{32S^2}{A}, \quad \frac{32S^2}{A}.$$

All are at most $32BS^2/A^2$, and $|\Delta|^2/|H| \le 64BS^3/A$. After row division the residual is therefore at most

$$\frac{10}{81}\frac{\eta^{1/2}}{2AS}\left(\frac{27BS^3}{AC^3\eta^3}\right)^2. \tag{4.14}$$

Since a_j divides R_{ij} for $j \ge 1$ and $\det R \ne 0$, the preceding integral-form argument applies, now using (4.13) and (4.14). Finally

$$G \le \frac{16384}{27}\frac{4005}{4000} < 608, \qquad p^2e(S)^2 \le \frac{25 \cdot 1001 \cdot 4005}{64000 \cdot 999} < \frac{16}{9},$$

$$27\eta^{-3} \le 27(1000/999)^3 < 28, \qquad \eta^{1/2} < 1.$$

These strict bounds yield (4.10). If (4.11) held, then $CD/2 < Z^2 < 2CD$ would make each term in (4.10) less than $1/2$, a contradiction. □

5 Finite integral approximation forms

The interval arguments below begin with a structural lower bound and end with the two-entry estimates at every order. The finite constructions in this section supply the intervening forbidden intervals. We give finite formulas for every row used here and prove the required divisibility in the polynomial ring over $\mathbb{Z}$ before evaluating the tuple entries.

5.1 From polynomial rows to inequalities

Let $x_1, \dots, x_h, c, d$ be distinct nonzero entries of a $D(1)$-tuple in the full ring of integers $\mathcal{O}_K$ of an imaginary quadratic field. Put $X_j = |x_j|$, $C = |c|$, $D = |d|$, $X = \min_j X_j$, $Y = \max_j X_j$, and choose $z \in \mathcal{O}_K$ with $z^2 = cd + 1$. Suppose

$$X \geq 1, \qquad XC \geq 1000, \qquad C \geq 4Y, \qquad D \geq C. \tag{5.1}$$

Then $C^2 \geq 4000$, so $z \neq 0$. Write $Z = |z|$ and let $\theta_x = (1 + 1/(xc))^{1/2}$ be the binomial branch. If $r_x^2 = xc + 1$ and $v_x^2 = xd + 1$, signs can be chosen, independently for the fixed entries but consistently across the rows, so that

$$\left|\theta_x - \frac{\pm r_x v_x}{xz}\right| \leq \frac{C}{|x|Z^2}\sqrt{1 + \frac{1}{|x|C}}\left(1 + \frac{|x|}{C}\right) < \frac{eC}{|x|Z^2}. \tag{5.2}$$

Here $e = 3/2$ is valid under (5.1), and $e = 101/100$ is valid if $C \geq 200Y$. Indeed, the square of the quotient in (5.2) is $\theta_x^2(1 + (c - x)/(xz^2))$. Choose a square root w of the last factor with $\operatorname{Re} w \geq 0$. Then $|w + 1| \geq 1$ and an allowed sign gives the quotient $\theta_x w$. The assertion follows from $|w - 1| \leq |(c - x)/(xz^2)|$. For the sharper constant use

$$\frac{1001}{1000}\left(\frac{201}{200}\right)^2 < \left(\frac{101}{100}\right)^2.$$

Neither r_x nor v_x is divided out.

Suppose a polynomial row has entries $(Q_0, Q_{x_1}, \dots, Q_{x_h})$ with integral coefficients and $Q_x = xP_x$, where P_x is also integral. Its arithmetic form

$$\Lambda = Q_0 z + \sum_x P_x(\pm r_x v_x) \in \mathcal{O}_K$$

satisfies

$$|\Lambda| \leq Z\left|Q_0 + \sum_x Q_x \theta_x\right| + \frac{eC}{Z}\sum_x |P_x|. \tag{5.3}$$

A matrix of such rows with rank $h + 1$ supplies a nonzero Λ, since its common evaluation vector has first coordinate $z \neq 0$. For this row $|\Lambda| \geq 1$, because the norm of a nonzero algebraic integer in K is a positive rational integer. All the estimates below enlarge a coefficient strictly and therefore give a strict inequality $1 < U/Z + VZ$.

We will use the following conversion repeatedly. Set $\varepsilon = 1/4000$. Since $D \geq C$ and $C^2 \geq 4000$,

$$(1 - \varepsilon)CD \leq Z^2 \leq (1 + \varepsilon)CD.$$

Consequently $1 < U/Z + VZ$ excludes the closed interval

$$\frac{4U^2}{C(1 - \varepsilon)} \leq D \leq \frac{1}{4V^2C(1 + \varepsilon)}. \tag{5.4}$$

An interval with its lower endpoint above its upper endpoint is simply empty. Each endpoint in a nonempty interval makes the corresponding summand at most $1/2$, which explains both closed endpoints.

5.2 Five families with two fixed entries

Let x, y, c, d be distinct nonzero tuple entries, with

$$1 \le X = |x| \le Y = |y|, \qquad XC \ge 1000, \qquad C \ge 200Y, \qquad D \ge C. \tag{5.5}$$

The parameters for the five families are in Table 1. Here $P = m + 2k$. In this subsection, Δ and J denote constants in the determinant formulas. The column R_* gives a uniform residual bound. Smaller bounds depending on $|x - y|$ are derived below.

family	k	m	g	Δ	J	β	R_*
N_2	2	2	2	-1800	$-15/64$	$5/1024$	$5/4$
N_3	3	3	4	235200	$105/256$	$35/65536$	$35/32$
A_3	3	2	4	-156800	$-35/256$	$35/32768$	$35/16$
A_4	4	3	2	23284800	$3465/16384$	$63/524288$	$63/8$
A_5	5	4	4	-3995671680	$-27027/65536$	$495/33554432$	$495/64$

Table 1: Parameters and exact constants of the finite two-anchor rows.

Here is an explicit finite construction of all their coefficients. Put $p = (m, k, k)$ and $\delta_{ij} = 1$ if $i = j$, and 0 otherwise. For $i, j \in \{0, 1, 2\}$, sum over nonnegative triples $h = (h_0, h_1, h_2)$ with $h_0 + h_1 + h_2 = p_j + \delta_{ij} - 1$. Set

$$b_{ijh} = 4^k \binom{k + 1/2}{h_j} \prod_{\ell \ne j} \binom{-p_\ell - \delta_{i\ell}}{h_\ell}$$

and define F_{ij} as the sum of $b_{ijh} T_{ijh}$, where

$$\begin{aligned} T_{i0h} &= (-1)^{2k+\delta_{i1}+\delta_{i2}+h_1+h_2} x^{\delta_{i1}+h_1} y^{\delta_{i2}+h_2} c^{k-h_0} (x-y)^{2k}, \\ T_{i1h} &= (-1)^{k+\delta_{i2}+h_2} x^m y^{\delta_{i2}+h_2} (xc+1)^{k-h_1} (x-y)^{k-\delta_{i2}-h_2}, \\ T_{i2h} &= y^m x^{\delta_{i1}+h_1} (yc+1)^{k-h_2} (x-y)^{k-\delta_{i1}-h_1}. \end{aligned} \tag{5.6}$$

Define $\widetilde{Q}$ by dividing row 0 of F by $x - y$, row 1 by x, and row 2 by y. The integral matrix used below is $Q = \widetilde{Q}/g$.

All exponents in (5.6) are nonnegative. The row divisions hold term by term. For example, in row 0, column 1, $h_2 \le k - 1$, so its difference exponent is positive. The same argument applies to column 2. Row 1 has a factor x in every column, and row 2 a factor y. Moreover, after these divisions, column 1 still has a factor x and column 2 still has a factor y, since $m \ge 2$. Finally the numbers $4^k \binom{k+1/2}{h}$ for $0 \le h \le k$ are

k	$4^k \binom{k+1/2}{h}, \quad h = 0, \dots, k$
2	$16, 40, 30$
3	$64, 224, 280, 140$
4	$256, 1152, 2016, 1680, 630$
5	$1024, 5632, 12672, 14784, 9240, 2772.$

They are divisible by the listed g, whereas all the other binomial coefficients in b_{ijh} are integers. Thus Q has integral coefficients, and its columns 1 and 2 are divisible by x and y, respectively. These conclusions hold before any specialization of x, y, c.

We next determine the order of the residual and prove that the matrices have rank three. Put $\alpha = (0, y, x)$, $N = xyc$. The residue sum

$$I_i = \frac{1}{2\pi i} \int_\Gamma \frac{(1 + w/N)^{k+1/2}\, dw}{(w - \alpha_i) \prod_{\ell=0}^{2} (w - \alpha_\ell)^{p_\ell}} \tag{5.7}$$

uses a positively oriented contour around the three poles, excluding the cut $w = -N(1+t)$, $t \geq 0$. Their distance from this cut is positive because $|\alpha_j/N| \leq 1/(XC) < 1$ for $j = 1, 2$. Leibniz's rule for the three pole residues gives exactly (5.6), and hence

$$4^k c^k x^{m+k} y^{m+k} (x-y)^{2k} I_i = F_{i0} + F_{i1}\theta_x + F_{i2}\theta_y.$$

The estimate below shows that these residuals are $O(c^{-(P-k)})$. Replacing the column indexed by 0 of $\widetilde{Q}$ by its residual therefore shows that its polynomial determinant has degree at most $2k - (P-k) = k - m$ in c. The entry in row i and column 0 is divisible by $c^{k-m+1-\delta_{i0}}$, directly from the finite sum, so the determinant is divisible by c^{k-m}. The coefficient of c^{k-m} in $\det \widetilde{Q}$ is therefore $4^k \binom{k+1/2}{m} (x-y)^{2k-1}$ times the determinant of the constant root block in rows $1, 2$.

To compute this block, put $\lambda = k + 1/2 - m$ and

$$A_{k,\lambda}(x,y) = \sum_{h=0}^{k} \binom{-k}{h} \binom{\lambda}{k-h} x^h (x-y)^{k-h},$$
$$B_{k,\lambda}(x,y) = \sum_{h=0}^{k-1} \binom{-k-1}{h} \binom{\lambda}{k-1-h} x^h (x-y)^{k-1-h}.$$

Vandermonde's convolution over the other two indices of (5.6) gives that block as

$$4^k \begin{pmatrix} x^{m-1} A_{k,\lambda}(y,x) & y^m B_{k,\lambda}(x,y) \\ x^m B_{k,\lambda}(y,x) & y^{m-1} A_{k,\lambda}(x,y) \end{pmatrix}.$$

For the five parameter pairs in Table 1,

$$A_{k,\lambda}(x,y) A_{k,\lambda}(y,x) - xy B_{k,\lambda}(x,y) B_{k,\lambda}(y,x) = J(x-y)^{2k}, \tag{5.8}$$

where $J = (-1)^k \binom{\lambda}{k} \binom{\lambda-k}{k}$ is printed in the table. To compare the coefficients in this identity, put

$$a_j = (-1)^j \binom{\lambda}{j} \binom{\lambda-k-j}{k-j}, \quad b_j = (-1)^j \binom{\lambda}{j} \binom{\lambda-k-1-j}{k-1-j},$$

with $a_j = 0$ outside $0 \leq j \leq k$ and $b_j = 0$ outside $0 \leq j < k$. Expanding the preceding sums by Vandermonde gives, for $r = 0, \ldots, 2k$,

$$\sum_{j=0}^{k} (a_j a_{k-r+j} - b_j b_{k-r+j}) = J(-1)^r \binom{2k}{r}.$$

Substitution of $(k, \lambda) = (2, 1/2), (3, 1/2), (3, 3/2), (4, 3/2), (5, 3/2)$ verifies every coefficient of (5.8). It follows that

$$\det \widetilde{Q} = \Delta (xy)^{m-1} c^{k-m} (x-y)^{4k-1}, \qquad \Delta = 4^k \binom{k+1/2}{m} 16^k J.$$

In particular Q has rank three at every admissible evaluation.

To bound the residual, deform (5.7) to its cut. The large-circle contribution vanishes, since $P > k + 1/2$, and the difference of the two boundary values of the numerator has modulus $2t^{k+1/2}$. With $\eta = 1 - 1/(XC)$ one obtains

$$|I_i| \leq \frac{|N|^{-P}}{\pi} \int_0^\infty \frac{t^{k+1/2}\, dt}{(t+\eta)^{P+1}} = \beta \eta^{-(P-k-1/2)} |N|^{-P},$$

where substitution $t = \eta u/(1-u)$ and the elementary half-integer beta integral give

$$\beta = \frac{(2k+2)!(2P-2k-2)!}{4^P(k+1)!(P-k-1)!P!}.$$

The five values are in the table. Their largest exponent $P-k-1/2$ is $17/2$, and $(1000/999)^{17} < (101/100)^2$. Hence $\eta^{-(P-k-1/2)}$ is strictly below $101/100$.

Put $u = X/Y$ and $\delta = |x-y|/Y$. After the row divisions and division by g, each residual is strictly less than

$$R(x,y)\frac{Y^k}{X^{k+1}C^{P-k}}, \qquad R(x,y) = \frac{101\,4^k\beta}{100g}\max(\delta^{2k}, u\delta^{2k-1}). \tag{5.9}$$

Indeed, the three extra row factors are respectively $u\delta^{2k-1}, \delta^{2k}, u\delta^{2k}$, and $u \le 1$. Equivalently the coefficient in front of the maximum is $R_*101/(200\,2^{2k})$, since $R_* = 2\,16^k\beta/g$. If only $Y \ge qX$ with $q \ge 1$ is known, one may use

$$R(q) = \frac{101\,4^k\beta}{100g}(1+1/q)^{2k}. \tag{5.10}$$

In particular $R(1) = 101R_*/200 < R_*$, so every listed R_* is valid.

For completeness we give all data needed for the coefficient bounds. Write $P_{i1} = \widetilde{Q}_{i1}/x$ and $P_{i2} = \widetilde{Q}_{i2}/y$ before division by g. Every monomial $x^r y^s c^\ell$ in these polynomials satisfies

$$r, s \ge 0, \qquad 0 \le \ell \le k, \qquad r+s = \ell + P - k - 2. \tag{5.11}$$

This follows term by term from (5.6). Let $L_{i\ell}$ be the sum of absolute integer coefficients of c^ℓ in P_{i1}, P_{i2} together. The following table is obtained by expanding the displayed finite sums. Rows 1 and 2 have the same sums.

family	i	$(L_{i\ell})_{\ell=0}^k$
N_2	0	96, 352, 256
	1, 2	100, 400, 480
N_3	0	1344, 9408, 17024, 8960
	1, 2	1368, 8384, 23168, 16384
A_3	0	800, 4544, 10368, 6144
	1, 2	1288, 4256, 9856, 8960
A_4	0	10944, 86784, 350208, 482304, 215040
	1, 2	18128, 80512, 276096, 561920, 327680
A_5	0	156000, 1601280, 9707520, 22379520, 22087680, 7864320
	1, 2	262152, 1463264, 6707712, 22437888, 28452864, 12300288

If M is any constant strictly larger than

$$\frac{101}{100g}\max_i \sum_{\ell=0}^{k} L_{i\ell}1000^{\ell-k}, \tag{5.12}$$

then (5.3) proves

$$1 < \frac{MY^{P-2}C^{k+1}}{Z} + \frac{RY^kZ}{X^{k+1}C^{P-k}},$$

where R can be chosen from (5.9), (5.10), or the table. To see (5.12), divide a monomial in (5.11) by $Y^{P-2}C^k$. Its modulus is $(X/Y)^r(YC)^{\ell-k} \le 1000^{\ell-k}$. Valid integer choices of M in the table order are

$$243, \quad 4143, \quad 2265, \quad 165763, \quad 3113013.$$

The choices $735, 4143, 2265, 166000, 26859063$, respectively, are therefore also valid. The endpoints supplied by (5.4), before division by $1 \mp \varepsilon$, are

$$l = 4M^2Y^{2P-4}C^{2k+1}, \qquad u = \frac{X^{2k+2}C^{2P-2k-1}}{4R^2Y^{2k}}.$$

We shall also use two refinements of the coefficient bound. If $Y \geq qX$, replace the summand corresponding to $nx^ry^sc^\ell$ in (5.12) by $|n|q^{-r}1000^{\ell-k}$. The same proof applies. For fixed values of x and y, first combine the coefficients algebraically and write $P_{ij}(x,y,c) = \sum_{\ell=0}^{k} p_{ij\ell}(x,y)c^\ell$. For any established lower bound $C \geq C_0 > 0$, it suffices to take

$$M > \frac{101}{100gY^{P-2}} \max_i \sum_{j=1}^{2} \sum_{\ell=0}^{k} |p_{ij\ell}(x,y)| C_0^{\ell-k}. \tag{5.13}$$

If x is a unit, (5.5) permits $C_0 = \max(1000, 200Y)$. Every modulus in (5.13) is the square root of the exact integer norm of an explicitly specified algebraic integer. Each coefficient modulus can therefore be bounded above by a rational number by comparing the square of that number with the corresponding integral norm. In the field basis $\omega^2 - \tau\omega + \nu = 0$, the required norm is $N(r + s\omega) = r^2 + \tau rs + \nu s^2$.

5.3 Quadratic rows with error of order c^{-5}

Now let a, b, s, c, d be distinct nonzero tuple entries, with ordered moduli $1 \leq A \leq B \leq S \leq C \leq D$, and assume

$$C \geq 200S, \qquad AC \geq 1000, \qquad C \geq 1000.$$

For an ordered pair (u, v) chosen from a, b, s, define rows R_{uv}, T_{uv} as follows. Their other root columns vanish, $Q_t = tP_t$, and $P_t = P_{t,0} + P_{t,1}c + P_{t,2}c^2$. The complete coefficient data are

$$\begin{aligned}
R: \quad & P_{u,0} = 7u^2 - 35uv, && P_{v,0} = 35uv - 7v^2,\\
& P_{u,1} = 64u^3 - 140u^2v, && P_{v,1} = 140uv^2 - 64v^3,\\
& P_{u,2} = 72u^4 - 120u^3v, && P_{v,2} = 120uv^3 - 72v^4;\\
T: \quad & P_{u,0} = -15u^3 + 56u^2v - 105uv^2, && P_{v,0} = 70uv^2 - 6v^3,\\
& P_{u,1} = -180u^4 + 512u^3v - 420u^2v^2, && P_{v,1} = 100uv^3 - 12v^4,\\
& P_{u,2} = -240u^5 + 576u^4v - 360u^3v^2, && P_{v,2} = 24v^5.
\end{aligned}$$

In both rows put

$$Q_0 = -\sum_{t=u,v} \left[tP_{t,2}c^2 + (tP_{t,1} + P_{t,2}/2)c + tP_{t,0} + P_{t,1}/2 - P_{t,2}/(8t) \right].$$

This is an integral polynomial because $P_{t,1}$ has even coefficients and $P_{t,2}$ is divisible by $8t$.

With $b_n = \binom{1/2}{n}$, the coefficient of c^{-n} after removing the polynomial part is

$$\sum_{t=u,v} \left(b_n t^{1-n} P_{t,0} + b_{n+1} t^{-n} P_{t,1} + b_{n+2} t^{-n-1} P_{t,2} \right).$$

Substituting the displayed polynomials gives zero for $n = 1, 2, 3, 4$. The coefficients at c^{-5} are respectively

$$E_R(u,v) = \frac{5}{128}(u^{-2} - v^{-2}) + \frac{5}{256}(uv^{-3} - vu^{-3}), \tag{5.14}$$
$$E_T(u,v) = -\frac{75}{128u} + \frac{5v}{16u^2} - \frac{15v^2}{256u^3} - \frac{35u}{256v^2} + \frac{15}{32v}.$$

These identities and the bound $|b_n| \le 21/1024$ for $n \ge 6$ control the remaining series.

Use the four rows

$$R_{ab}, \quad H = R_{ab} + R_{bs} - R_{as}, \quad T_{ab}, \quad V = T_{as} - T_{bs}. \tag{5.15}$$

In columns (Q_0, Q_a, Q_b, Q_s), their determinant is

$$-78750a^2b^2s^2(a-b)^5(a-s)(b-s)F_s,$$

where

$$F_s = 7(2s - a - b) + 6c[3s^2 - (a+b)s - ab].$$

Replacing column zero by its $O(c^{-5})$ residual shows that the polynomial determinant has degree at most $2 + 2 + 2 - 5 = 1$ in c. Expanding its constant and linear coefficients from the displayed rows gives the identity.

Suppose that $F_s = 0$. If $3s^2 - (a+b)s - ab \ne 0$, this algebraic integer has modulus at least one, and

$$6C \le 7|2s - a - b| \le 28S,$$

contrary to $C \ge 200S$. Otherwise $a+b = 2s$ and $ab = s^2$, so $a = b = s$, again a contradiction. Thus the matrix has rank four.

Put $H_* = \max(BS^3, B^3S^2)$. The coefficient sums by increasing c degree in the root columns after dividing by their anchors are

R_{ab}	84	408	384
H	210	840	720
T_{ab}	252	1224	1200
V	492	2424	2352.

Every anchor monomial in these columns is at most H_*. For H, the three coefficients at degree ℓ are $k_\ell a^{\ell+1}(s-b)$, $k_\ell b^{\ell+1}(a-s)$, $k_\ell s^{\ell+1}(b-a)$, where $(k_0, k_1, k_2) = (35, 140, 120)$. For V, the coefficient obtained after dividing the last column by s is $P_s = 70(a-b)s^2 + 100(a-b)s^3c$. The monomials in P_a and P_b are bounded by B^3S^2. The remaining pair rows have monomials at most B^4 and B^5, respectively. These bounds use only $1 \le A \le B \le S$ and remain valid when some moduli are equal. Thus the second term in (5.3) is bounded above by

$$\frac{101}{100}\max_i(L_{i2} + L_{i1}/1000 + L_{i0}/1000^2) < 2400$$

times H_*C^3/Z.

We now retain the ratio B/A in the residual bound. Set $\tau = 21/(1024 \cdot 999)$, and define the polynomials

$$\begin{aligned}
256L_{R_{ab}}(x) &= 5x + 10x^2 + 10x^4 + 5x^5,\\
W_{R_{ab}}(x) &= 295x + 143x^2 + 143x^4 + 295x^5,\\
256L_H(x) &= 10(x + x^4 + x^5),\\
W_H(x) &= 590(x + x^4 + x^5),\\
256L_{T_{ab}}(x) &= 15 + 80x + 150x^2 + 120x^3 + 35x^4,\\
W_{T_{ab}}(x) &= 885 + 1144x + 435x^2 + 42x^3 + 170x^4,\\
256L_V(x) &= 15 + 80x + 150x^2 + 280x^3 + 35x^4,\\
W_V(x) &= 885 + 1144x + 435x^2 + 2634x^3 + 170x^4.
\end{aligned}$$

For $q \geq 1$ put

$$\mathcal{R}(q) = \max_i \{L_i(1/q) + \tau W_i(1/q)\}. \tag{5.16}$$

To verify the table, combine the leading coefficients (5.14) according to (5.15) before taking absolute values. For a Laurent monomial $a^i b^j s^k$, division of its modulus by S^2/A^3 gives, on setting $u = A/B$ and $v = B/S$,

$$A^{i+j+k+1} u^{2-j-k} v^{2-k}.$$

Every monomial from those combined leading coefficients satisfies $i + j + k + 1 \leq 0$, $2 - j - k \geq 0$, $2 - k \geq 0$. Summing its absolute coefficient times x^{2-j-k} gives $L_i(x)$. For the tail, a monomial from the coefficient $P_{t,\ell}$ has the exponent of the corresponding fixed entry decreased by $4 + \ell$. The same three inequalities hold. Use the divided root-column polynomials of (5.15), with the combinations defining H and V already performed. Sum the absolute values of their integer coefficients with weight x^{2-j-k}, before multiplication by the binomial coefficients. This gives $W_i(x)$. Each later inverse power brings a further factor at most $1/(AC)$. Since $|b_n| \leq 21/1024$ for $n \geq 6$ and $\sum_{r \geq 1} (AC)^{-r} \leq 1/999$, the whole tail contributes at most $\tau W_i(1/q)$. This proves (5.16).

For every $q \geq 1$ with $B \geq qA$, we conclude that

$$1 < \frac{2400 H_* C^3}{Z} + \frac{\mathcal{R}(q) S^2 Z}{A^3 C^5}. \tag{5.17}$$

For $0 \leq x \leq 1$, the displayed polynomials satisfy $L_i(x) \leq L_V(x)$ and $W_i(x) \leq W_V(x)$ for every row i. These comparisons follow from $x^5 \leq x^4 \leq x^3$. Consequently $\mathcal{R}(q) = L_V(1/q) + \tau W_V(1/q)$ for all $q \geq 1$, and this function is nonincreasing. In particular,

$$\mathcal{R}(1) = \frac{35}{16} + \tau\, 5268 < 3.$$

Equation (5.17), together with (5.4), gives its closed forbidden interval.

5.4 Five linear rows on four fixed entries

Let a, b, s, t, c, d be six distinct nonzero tuple entries, with $1 \leq A \leq B \leq S \leq T \leq C \leq D$, and suppose

$$C \geq 4T, \qquad AC \geq 1000.$$

For an ordered pair (u,v) chosen from a,b,s,t define a row $\mathcal{L}_{uv}$ by

$$\begin{aligned} P_u &= u^2 - 5uv + (4u^3 - 8u^2v)c, \\ P_v &= 5uv - v^2 + (8uv^2 - 4v^3)c, \\ Q_u &= uP_u, \qquad Q_v = vP_v, \qquad Q_0 = -(u-v)^3[3 + 4(u+v)c], \end{aligned}$$

and put the other root columns equal to zero. For $P_x = P_{x,0} + P_{x,1}c$, the coefficient of c^{-n} in the residual is

$$b_n \sum_{x=u,v} x^{1-n} P_{x,0} + b_{n+1} \sum_{x=u,v} x^{-n} P_{x,1}.$$

The two sums for $n = 1, 2, 3$ are, respectively,

$$\begin{array}{c|cc} 1 & u^2 - v^2 & 4(u^2 - v^2) \\ 2 & 6(u-v) & 12(u-v) \\ 3 & 5(u/v - v/u) & 8(u/v - v/u). \end{array}$$

After multiplication by b_n and b_{n+1}, the two entries in each row sum to zero. The coefficient Q_0 cancels the two nonnegative powers. The five rows $\mathcal{L}_{ab}, \mathcal{L}_{as}, \mathcal{L}_{bs}, \mathcal{L}_{at}, \mathcal{L}_{bt}$, in columns $(Q_0, Q_a, Q_b, Q_s, Q_t)$, have determinant

$$180a^2b^2s^2t^2(a-b)^2(a-s)(a-t)(b-s)(b-t)(s-t). \tag{5.18}$$

Indeed, the column indexed by 0, after replacement by the residual, is $O(c^{-4})$ and the remaining four columns have degree at most one, so their polynomial determinant is constant in c. Expanding its constant term from the displayed rows gives (5.18). Every factor is nonzero, giving rank five without an additional assumption on the moduli.

The sum of absolute integer coefficients of the P columns in each row is 36. Each coefficient monomial, including its power of c, is bounded by T^3C, so $\sum_x |P_x| \le 36T^3C$. After the cancellations, the first residual monomials are $u^{-1}, v/u^2, u/v^2, v^{-1}$, each bounded by T/A^2. Since $|b_n| \le 5/128$ for $n \ge 4$, the entire convergent residual is at most

$$\frac{36 \cdot 5}{128(1 - 1/(AC))} \frac{T}{A^2C^4} \le \frac{625}{444} \frac{T}{A^2C^4} < \frac{3T}{2A^2C^4}.$$

Apply (5.3) with $e = 3/2$ to a nonzero form and enlarge 54 to 55. This gives

$$1 < \frac{55T^3C^2}{Z} + \frac{3TZ}{2A^2C^4}.$$

Its endpoints before applying the factors in (5.4) are $l = 4 \cdot 55^2T^6C^3$ and $u = A^4C^7/(9T^2)$, to be multiplied by $1/(1-\varepsilon)$ and $1/(1+\varepsilon)$, respectively.

6 Entries of modulus at least three

Proposition 6.1. *A $D(1)$-tuple contains at most five entries of modulus at least three.*

Proof. Suppose six such entries are ordered as a, b, s, t, c, d, with $3 \le A \le B \le S \le T \le C \le D$. Lemma 2.5 gives $T > 3ABS$ and hence $AT > 243$. We may therefore apply the row $H = 48, \lambda = 19/200$ of Lemma 3.7. It gives all the required size conditions and

$$C > \frac{A^2T^5}{S^2}, \qquad D > \mathcal{G} := \frac{A^2C^7}{S^2T^4}. \tag{6.1}$$

Put

$$\mathcal{Q} = \frac{|(a-b)(a-s)(b-s)|}{BS^2}, \qquad c_* = \frac{9999}{10000}\left(\frac{200}{19}\right)^2.$$

The retained inequality and $T > 3ABS$ give

$$C > 3^5 c_* \left(\frac{1}{\mathcal{Q}} - \frac{\mu}{A^2B^2}\right)^2 A^7B^5S^3.$$

Since $A^2B^2 \geq 81$ and $0 < \mathcal{Q} \leq 8$,

$$3^5 c_* \left(\frac{1}{8} - \frac{\mu}{81}\right)^2 = \frac{16405041495117}{40000000000} > 400.$$

Consequently, with $K = 400$,

$$C > KA^7B^5S^3. \tag{6.2}$$

The approximation intervals. Use the integral approximation constructions with the constants displayed below. Lemma 3.7 gives $C > 1000$, $C > 200S$, $C > 4T$, $AC > 1000$ and $D > 1000C$. In particular, the two-entry construction in Proposition 4.2 applies because $C > 4B$ and $AC > 1000$. Put

$$\epsilon = \frac{1}{4000}, \quad h = \frac{4001}{3999}.$$

$$M_n = 4143,\ R_n = \frac{35}{32},\ M_3 = 2265,\ R_3 = \frac{35}{16},\ M_4 = 166000,\ R_4 = \frac{63}{8},$$

$$M_5 = 26859063,\ R_5 = \frac{495}{64}, \qquad M_m = \frac{8}{9}608^2, \quad R_m = \frac{3920}{81}, \quad R_{bs} = \frac{35}{32}.$$

For the finite constructions, these constants bound the coefficients and residuals in their stated integral normalizations. In the table, P denotes the four quadratic rows in (5.15), Q the five linear rows with determinant (5.18), and M the estimate (4.10). For P, use $\mathcal{R}(1) < 3$ and put $H_P = \max(BS^3, B^3S^2)$. The row $N_{3,bs}$ uses N_3 with fixed entries b, s, and O_j uses (4.7) at $k = j$. Each pair (l, u) denotes the forbidden closed interval $[l/(1-\epsilon), u/(1+\epsilon)]$ for D, by (5.4).

row	l	u
Q	$4 \cdot 55^2T^6C^3$	$A^4C^7/(9T^2)$
N_2	$4 \cdot 735^2B^8C^5$	$4A^6C^7/(25B^4)$
P	$4 \cdot 2400^2H_P^2C^5$	$A^6C^9/(4 \cdot 3^2S^4)$
A_3	$4M_3^2B^{12}C^7$	$A^8C^9/(4R_3^2B^6)$
M	$4M_m^2B^6S^{16}C^5$	$A^6C^{11}/(4R_m^2B^4S^{10})$
N_3	$4M_n^2B^{14}C^7$	$A^8C^{11}/(4R_n^2B^6)$
$N_{3,bs}$	$4M_n^2S^{14}C^7$	$B^8C^{11}/(4R_{bs}^2S^6)$
A_4	$4M_4^2B^{18}C^9$	$A^{10}C^{13}/(4R_4^2B^8)$
A_5	$4M_5^2B^{24}C^{11}$	$A^{12}C^{17}/(4R_5^2B^{10})$
O_j	$36(42B^3C)^{2j}C/B^2$	$(2AC^2/(5B))^{2j}/C$

The last row is available for every positive integer j.

Overlap of consecutive intervals. The first comparison puts $\mathcal{G}$ above the lower endpoint of Q, so (6.1) places D above that endpoint. Each later comparison requires the preceding upper endpoint to exceed the next lower endpoint, with the factors $1 \mp \epsilon$ included. The table gives sufficient strict comparisons, so the lower quantity must exceed the required constant. The first two use $C > A^2T^5/S^2$ and $T > 2ABS$, and the remaining comparisons use (6.2).

junction and additional condition	lower quantity	required constant
$\mathcal{G} \to Q$	$1024A^{20}B^{10}$	$12100/(1-\epsilon)$
$Q \to N_2$	$256A^{16}S^4$	$36 \cdot 735^2 h$
$N_2 \to P,\ H_P = BS^3$	$K^2A^{20}B^4$	$25 \cdot 2400^2 h$
$N_2 \to P,\ H_P = B^3S^2$	$K^2A^{20}S^2$	$25 \cdot 2400^2 h$
$P \to N_3,\ S \geq B^2$	K^2A^{20}	$16 \cdot 3^2 M_n^2 h$
$P \to A_3,\ S \leq B^2$	K^2A^{20}	$16 \cdot 3^2 M_3^2 h$
$A_3 \to M,\ S \leq B^2$	K^4A^{36}	$16R_3^2M_m^2h$
$M \to N_3$	$K^4A^{34}B^2S^2$	$16R_m^2M_n^2h$
$N_3 \to A_4,\ S \geq B^3$	$K^2A^{22}B^4$	$16R_n^2M_4^2h$
$N_3 \to N_{3,bs},\ S \leq B^3$	$K^4A^{36}B^8$	$16R_n^2M_n^2h$
$N_{3,bs} \to A_4$	K^2A^{14}	$16R_{bs}^2M_4^2h$
$A_4 \to O_4$	$K^4A^{38}B^2$	$144R_4^2\,42^8h$
$O_4 \to A_5$	K^4A^{36}	$4M_5^2(5/2)^8h$
$A_5 \to O_5$	$K^6A^{54}B^{10}$	$144R_5^2\,42^{10}h$
$O_5 \to O_6$	$K^6A^{52}B^4$	$36 \cdot 42^{12}(5/2)^{10}h$

Every remaining modulus power is nonnegative. Thus the displayed comparisons follow by substituting $A = B = S = 3$, $K = 400$ and clearing positive denominators. To make the tight comparison explicit, its exact quotient is

$$\frac{K^2 3^{14}}{16R_{bs}^2M_4^2h} = \frac{1224133953984}{844113475625} > \frac{7}{5}.$$

All other quotients exceed $13/10$.

Here are the algebraic reductions behind the branch conditions. The two P incoming comparisons handle both terms of its maximum. The $P \to N_3$ comparison uses $S \geq B^2$. The alternative $P \to A_3$ comparison only needs $S \geq B$. For $A_3 \to M$, the residual modulus factor is $B^8/S^4 \geq 1$ when $S \leq B^2$. For $N_3 \to A_4$, it is $S^6/B^{14} \geq B^4$ when $S \geq B^3$. For $N_3 \to N_{3,bs}$, it is $B^{14}/S^2 \geq B^8$ when $S \leq B^3$. The last four reductions use only $S \geq B$.

For $N_{3,bs} \to A_4$, the endpoint quotient is

$$\frac{C^2}{16M_4^2R_{bs}^2h\,B^{10}S^6}.$$

Substituting (6.2) gives the corresponding table entry.

Exclusion of all remaining values of D. Start with $\mathcal{G}, Q, N_2, P$. If $S \geq B^2$, continue through N_3. Otherwise continue through A_3, M, N_3. If $S \geq B^3$, continue to A_4. Otherwise use $N_{3,bs}, A_4$. Equality belongs to either applicable branch. Finish with

$$A_4, O_4, A_5, O_5, O_6, \ldots.$$

Put $H_2 = 42B^3C,\ R_2 = 2AC^2/(5B)$. Then

$$\frac{R_2}{H_2} = \frac{AC}{105B^4} > \frac{400A^8BS^3}{105} \geq \frac{400 \cdot 3^{12}}{105} > 1.$$

The ratio of the upper endpoint of O_j to the lower endpoint of O_{j+1} is multiplied by $(R_2/H_2)^2$ when j increases by one. The checked $O_5 \to O_6$ junction therefore proves every later one. Since $H_2 > 1$, the lower endpoints tend to infinity.

The starting inequality puts D above the lower endpoint of Q. Since D avoids the closed interval, it lies above its upper endpoint. The overlap then places D above the lower endpoint of the next interval. This implication is also valid when an intermediate interval is empty. It would force D beyond arbitrarily large endpoints, a contradiction. □

7 Tuples with at least two small entries

Call an entry small when its modulus is less than 3. The purpose of this section is to prove the following result.

Proposition 7.1. *A $D(1)$-tuple in the full ring of integers of an imaginary quadratic field with at least two small entries has at most five elements.*

The finite calculations below use exact integer algorithms, finite search domains and complete tables. Their completeness follows from a reversible descent. The search radius is proved below.

7.1 Auxiliary bounds

Lemma 7.2. *Let $a_1, a_2, a_3, b_1, b_2, b_3$ be six distinct members of a $D(1)$-tuple, separately ordered by modulus. Put $A_i = |a_i|$, $B_i = |b_i|$, and $W_i = A_iB_i$. If $W_1 \geq 9$, then*

$$W_3 > \frac{W_1^3 W_2}{7}.$$

Proof. Choose complex square roots $u_i^2 = a_i$, $v_j^2 = b_j$ and signs of the integral roots such that

$$r_{ij} = u_i v_j f(1/(a_i b_j)), \qquad f(z) = \sqrt{1+z}, \quad f(0) = 1.$$

We first show $D_3 = \det(r_{ij}) \neq 0$. No two rows are proportional. Indeed, squaring such a relation gives a linear polynomial vanishing at the three distinct b_j, forcing the corresponding a_i to coincide. If the determinant vanished, its third row would therefore be λ times the first plus μ times the second, with $\lambda\mu \neq 0$. Squaring this relation gives $2\lambda\mu r_{1j} r_{2j} = Pb_j + Q$. Squaring again at the three values b_j proves the polynomial identity

$$(PX+Q)^2 = 4\lambda^2\mu^2(a_1X+1)(a_2X+1).$$

The right side has two distinct simple roots, a contradiction. Thus D_3 is a nonzero algebraic integer and $|D_3| \geq 1$.

Write $c_k = \binom{1/2}{k}$. Cauchy–Binet for the power series gives

$$\det(f(1/(a_ib_j))) = \sum_{0 \leq k_0 < k_1 < k_2} c_{k_0} c_{k_1} c_{k_2} \det(a_i^{-k_j}) \det(b_i^{-k_j}).$$

For each term, rearrangement of the exponents gives

$$|\det(a_i^{-k_j})| \leq 6 \prod_{i=1}^{3} A_i^{-k_{3-i}}, \qquad |c_{k_0} c_{k_1} c_{k_2}| \leq |c_0 c_1 c_2| = 1/16.$$

The analogous estimate holds for $\det(b_i^{-k_j})$. Summing the successive nonnegative gaps between the k_j gives

$$\sum_{0\le k_0<k_1<k_2}\prod_{i=1}^{3} W_i^{-k_{3-i}} = \frac{W_1^{-2}W_2^{-1}}{(1-W_1^{-1})(1-(W_1W_2)^{-1})(1-(W_1W_2W_3)^{-1})}.$$

This convergent geometric sum also justifies passage from truncated series to Cauchy–Binet. Since $W_i \ge 9$, the denominator is greater than $1-(9^{-1}+9^{-2}+9^{-3}) > 7/8$. Hence

$$1 \le |D_3| < \frac{18}{7}\frac{W_3^{1/2}}{W_1^{3/2}W_2^{1/2}}.$$

Squaring proves the assertion because $49/324 > 1/7$. □

Lemma 7.3 (Four roots at order one)**.** *Let a_1, a_2, a_3, a_4, c, d be distinct nonzero entries of a $D(1)$-tuple, with*

$$1 \le A = |a_1| \le B = |a_2| \le S = |a_3| \le T = |a_4|, \quad C = |c| > 1, \quad AC > 1.$$

Put $Z = |\sqrt{cd+1}|$, $\eta = 1 - 1/(AC)$, and use the function e from Lemma 4.1. Then

$$2 \le 8192e(T)\left(1+\frac{11}{8AC}\right)\frac{B^2S^4T^6C^2}{Z} + \frac{192BS^3T^5Z}{AC^4\eta^{7/2}}. \tag{7.1}$$

If $C \ge 4T$ and $AC \ge 1000$, then

$$2 < 10300\frac{B^2S^4T^6C^2}{Z} + 193\frac{BS^3T^5Z}{AC^4}. \tag{7.2}$$

If in addition $D = |d| \ge C$, the closed interval

$$2\cdot 10300^2 B^4S^8T^{12}C^3 \le D \le \frac{A^2C^7}{2\cdot 193^2B^2S^6T^{10}} \tag{7.3}$$

is impossible.

Proof. Use the order-one matrix in Lemma 4.1 with $m = 4$, and put

$$H = \prod_{i=1}^{4} a_i, \qquad \Delta = \prod_{1\le i<j\le 4}(a_i - a_j).$$

For $j \ge 1$, write

$$E_j = \frac{\Delta^2}{\prod_{\ell\ne j}(a_\ell - a_j)}, \qquad E_{ji} = \frac{E_j}{a_i - a_j} \quad (i \ne j).$$

These are polynomials. The simple and double residue formulas give

$$\begin{aligned} Q_{0j1} &= 4a_j^3(a_jc+1)E_j, \\ Q_{ij1} &= 4a_j^3(a_jc+1)a_iE_{ji} \quad (i \ge 1,\ i \ne j), \\ Q_{jj1} &= 6a_j^3E_j - Q_{0j1} - \sum_{\substack{i\ge 1\\ i\ne j}} Q_{ij1}. \end{aligned}$$

In this case (4.1) reads $\det Q_1 = -1440H^3\Delta^8 \neq 0$, and every column with index $j \geq 1$ is divisible by a_j^3.

Set $K = B^2S^4T^6$ and $L = C + 1/A$. Using $|a_i - a_j| \leq 2\max(A_i, A_j)$ for each anchor difference gives

$$A_j^3|E_j| \leq 2^9K, \qquad A_iA_j^3|E_{ji}| \leq 2^8K.$$

Indeed, the product of $\max(A_i, A_j)^2$ over all six pairs is K. In E_j, one factor is removed for each pair containing j, and their product is at least A_j^3. Forming E_{ji} removes one further factor $\max(A_i, A_j) \geq A_i$. Nine and eight difference factors remain, respectively, giving the stated powers of two. It follows that

$$\frac{|Q_{0j1}|}{A_j} \leq 2048LK, \qquad \frac{|Q_{ij1}|}{A_j} \leq 1024LK \quad (i \geq 1,\ i \neq j), \qquad 6A_j^2|E_j| \leq \frac{3072K}{A}.$$

Row zero has four terms. In a row with index $i \geq 1$, substituting the formula for Q_{ii1} gives one Q_{0i1} term and six off-diagonal terms, together with $6a_i^3E_i$. Thus every row satisfies

$$\sum_{j=1}^{4} \frac{|Q_{ij1}|}{A_j} \leq (8192C + 11264/A)B^2S^4T^6.$$

For $m = 4, k = 1$, the integral in the proof of (4.2) equals

$$\int_0^\infty \frac{x^{3/2}}{(x+\eta)^6}\,dx = \eta^{-7/2}B(5/2, 7/2) = \frac{3\pi}{256}\eta^{-7/2}.$$

Hence the residual is at most $3|\Delta|^2/(64|H|C^4\eta^{7/2})$. Since $|\Delta|^2 \leq 2^{12}B^2S^4T^6$ and $|H| = ABST$, this is at most $192BS^3T^5/(AC^4\eta^{7/2})$. Apply the final, even-coefficient assertion of Lemma 4.1 to obtain (7.1).

Under the additional size assumptions, $e(T)^2 \leq 25 \cdot 1001/16000$ and $\eta \geq 999/1000$. The exact rational comparisons

$$\left[8192\left(1 + \frac{11}{8000}\right)\right]^2 \frac{25 \cdot 1001}{16000} < 10300^2, \qquad 192^2\left(\frac{1000}{999}\right)^7 < 193^2$$

prove (7.2). Finally $CD/2 < Z^2 < 2CD$. Membership in (7.3) would therefore imply

$$Z > 10300B^2S^4T^6C^2, \qquad Z < \frac{AC^4}{193BS^3T^5}.$$

Both terms on the right of (7.2) would be less than one, a contradiction. □

7.2 Two small entries

We first bound the third large entry. The descent in Lemma 7.5 generates all common extensions within the resulting disk from a finite initial set.

Let a, b, s, t, p, q be six distinct tuple entries, with moduli $A \leq B \leq S \leq T \leq P \leq Q$. Assume

$$P \geq \max\{4T, 1000\}, \qquad Q \geq 4P, \qquad P^2 \geq \frac{32BST}{A^3}.$$

Lemma 3.5 gives a nonzero three-anchor form among the two triples sharing the first two anchors. Indeed, simultaneous vanishing would give

$$|pq + 1| \leq \frac{121BST}{4A^3} < P^2 - 1 \leq PQ - 1,$$

where the strict inequality follows from $BST/A^3 \geq 1$. Lemma 3.4 therefore gives

$$1 < \frac{5BT^2}{Z} + \frac{3TZ}{4AP^3}, \qquad Z = |\sqrt{pq+1}|.$$

Put $\varepsilon = 1/4000$. Since $P^2 \geq 4AP \geq 4000$, we have $(1-\varepsilon)PQ \leq Z^2 \leq (1+\varepsilon)PQ$. Consequently the closed interval

$$\frac{100B^2T^4}{(1-\varepsilon)P} \leq Q \leq \frac{4A^2P^5}{9(1+\varepsilon)T^2} \tag{7.4}$$

is impossible. We shall use the exact comparisons

$$\frac{100}{1-\varepsilon} < 101, \qquad \frac{4}{9(1+\varepsilon)} > \frac{2}{5}.$$

Lemma 7.4. *Suppose a $D(1)$-tuple has six entries with ordered moduli*

$$1 \leq A \leq B < 3 \leq S \leq T \leq C \leq D.$$

Then $C < 10^{550}$.

Proof. Lemmas 2.3 and 7.2 give

$$C > \frac{3}{8}AST, \qquad D > 2STC, \qquad D > \frac{A^3BT^3C}{7S}, \tag{7.5}$$

where the last inequality is used only when $AT \geq 9$. For the first inequality apply Lemma 2.3 to (a, s, t, c). In the last one the row entries are (a, b, s) and the column entries are (t, c, d).

Taking a, b as anchors and any of s, t, c with modulus $P \geq 10^8$ as the interpolation entry, Corollary 4.3 gives

$$\log_{10} D < g(\log_{10} P), \qquad g(x) = 7x + \frac{212}{5} + \frac{4047/25}{x-4}. \tag{7.6}$$

Indeed, $H_2 < 1134P$, $R_2 > (2/15)P^2$ and $253B^2 < 2277$. The parameters $(h, r, u) = (31/10, -9/10, 17/5)$ follow from $1134^{10} < 10^{31}$, $(15/2)^{10} < 10^9$ and $2277^5 < 10^{17}$. The function g is convex on $(4, \infty)$.

First assume $S \geq 10^8$. The first inequality in (7.5) implies $C \geq \max\{4T, 1000\}$, $AC \geq 1000$ and $C^2 \geq 32BST/A^3$. Moreover, $D > 18C$ and

$$DC > \frac{9}{448}A^5BST^5 > 101B^2T^4,$$

since $9A^5ST > 45248B$. Thus D lies above the lower endpoint of (7.4), and it must exceed the upper endpoint. Hence

$$D > \frac{2}{5}\frac{A^2C^5}{T^2} > \frac{243}{81920}A^7S^5T^3 > \frac{S^5T^3}{400} \geq \frac{S^8}{400}.$$

If $x = \log_{10} S \geq 49$, then $400 < 10^3$ gives $\log_{10} D > 8x - 3$. This contradicts (7.6), since $(8x-3) - g(x) = x - 227/5 - (4047/25)/(x-4)$ is increasing and equals $1/375$ at 49. Therefore $S < 10^{49}$. In the whole case $S \geq 10^8$, convexity gives $\log_{10} D < 389$, since $g(8) < 139$ and $g(49) = 389 - 1/375$.

Now assume $S < 10^8$ and $T \geq 10^{72}$. Use L_3 with anchors (a, b, s) and terminal pair (t, c). Lemma 3.3 applies. If $L_3 \neq 0$, its forbidden interval contains

$$\frac{101B^2S^4}{T} \leq C \leq \frac{A^2T^5}{37S^2}.$$

Here we used $100/(1 - 1/4000) < 101$ and $36(1 + 1/4000) < 37$. The lower endpoint is below one, so $C > T^5/(37S^2)$. Equation (7.5) gives $D > T^8/(259S^3) > T^8/10^{27}$, contrary to (7.6). Indeed, the function $x - 347/5 - (4047/25)/(x - 4)$ is increasing for $x \geq 72$ and positive at $x = 72$.

If $L_3 = 0$, we first prove $C > 4T$ in order to apply Lemma 3.4. The two terms in (3.1) satisfy

$$|1 - f(1/(tc))| > \frac{1}{3TC}, \qquad \frac{|F_3|}{ABS} < \frac{1}{3ABST^3}.$$

Their equality implies $C > ABST^2 > 4T$. Lemma 3.4 now gives $C > 5ABST^2 \geq 5ST^2$. The four-anchor nonvanishing hypotheses at C follow, and $DC > 2STC^2 > 50S^3T^5 > 101B^2T^4$. Thus (7.4) gives $D > (2/5)C^5/T^2 > 1250S^5T^8 > T^8$, again contradicting (7.6). Consequently $T < 10^{72}$. If $T \geq 10^8$, convexity and $g(72) < 549$ give $D < 10^{549}$.

It remains to treat $S, T < 10^8$. Put $M = 10^8$ and suppose $C \geq 10^{100}$. Four-anchor nonvanishing and Lemma 3.3 give

$$1 < \frac{15M^2}{Z} + \frac{3MZ}{C^3}, \qquad Z = |\sqrt{cd + 1}| > C/2.$$

The first term is less than $1/2$, so $D > C^5/(100M^2) > \max\{1000C, 17ABSTC^3\}$. The upper bound for D when $L_4 = 0$ therefore implies $L_4 \neq 0$. Lemma 3.6 gives

$$1 < \frac{27M^5}{Z} + \frac{4M^3Z}{5C^4}.$$

Again the first term is less than $1/2$. Hence

$$D > \frac{C^7}{100M^6} = 10^{-50}C^7.$$

Lemma 4.4 at order two, with anchors (a, b, s), forbids

$$\frac{512}{81}608^4B^6S^{16}C^5 \leq D \leq \frac{A^6C^{11}}{2 \cdot 28^4(10/81)^2B^4S^{10}}.$$

Since $(512/81)608^4 3^6 < 10^{15}$ and $2 \cdot 28^4(10/81)^2 3^4 < 10^7$, the smaller interval $10^{143}C^5 \leq D \leq 10^{-87}C^{11}$ is also forbidden. Its lower endpoint is less than $10^{-50}C^7$. Its upper endpoint exceeds the bound $D < 10^{60}C^7$ obtained from (7.6). This contradiction completes all cases. □

Lemma 7.5. *Let $a, b \in \mathcal{O}_K$ be distinct nonzero elements with $|a|, |b| < 3$, and suppose $r_0 \in \mathcal{O}_K$ satisfies $r_0^2 = ab + 1 \neq 0$. Every integral common extension z of a, b with $|z| \geq 33$ has an integral common extension w satisfying*

$$|w| < |z|/5 + 26 < |z|\,.$$

Here a common extension may be zero or equal to an anchor. The transformations used to obtain w are reversible within this enlarged set.

Proof. Choose roots $r^2 = az + 1$ and $s^2 = bz + 1$. The two transformations are

$$z_\pm = a + b + (2ab + 1)z \pm 2r_0 rs\,. \tag{7.7}$$

For the plus sign, the integral roots $r_+ = r_0 r + as$ and $s_+ = br + r_0 s$ satisfy $r_+^2 = az_+ + 1$ and $s_+^2 = bz_+ + 1$. Changing the sign of s proves the other case. Expansion also gives

$$a + b + (2ab + 1)z_+ - 2r_0 r_+ s_+ = z\,.$$

These identities involve no division by roots or extensions, so they remain valid when a root vanishes or a transformed value is zero or equals an anchor.

Put $p = ab$ and choose $\eta^2 = p$. The numbers

$$\lambda_\pm = 2p + 1 \pm 2r_0\eta$$

have product one. We show that the smaller modulus is less than $1/5$. If $|r_0|^2 \geq 2$, then $|2r_0^2 - 1| \geq 3$. Otherwise r_0 is a unit of modulus one, and its integral trace lies in $\{-2, -1, 0, 1, 2\}$. Trace ± 2 would give $r_0 = \pm 1$ and $ab = 0$. Trace zero gives $r_0^2 = -1$. If the trace is ± 1, then r_0^2 is a primitive cube root of unity. Thus in every permitted case $|2r_0^2 - 1|^2 \geq 7$.

Let $q \geq 1$ be the larger modulus of λ_+, λ_-. If $q \leq 5$, then

$$2\sqrt{7} \leq |\lambda_+ + \lambda_-| \leq q + q^{-1} \leq 5 + 1/5\,,$$

contrary to $28 > 676/25$. Choose the sign of η so that $|2p + 1 - 2r_0\eta| < 1/5$.

One of $\tau = rs, -rs$ satisfies $|\tau + \eta z| \geq |\eta z|$. Since $\tau^2 - (\eta z)^2 = (a + b)z + 1$, it follows that

$$|\tau - \eta z| \leq \frac{|a| + |b| + 1/|z|}{\sqrt{|ab|}}\,.$$

The corresponding value $w = a + b + (2p + 1)z - 2r_0\tau$ in (7.7) therefore satisfies

$$|w| < \frac{|z|}{5} + 6 + 14\sqrt{2} < \frac{|z|}{5} + 26\,,$$

using $|a| + |b| < 6$, $|ab| \geq 1$, $|z| \geq 1$, and $|r_0|/\sqrt{|ab|} \leq \sqrt{2}$. For $|z| \geq 33$ this is strictly smaller than $|z|$. □

Repeated application of Lemma 7.5 terminates, since squared moduli of algebraic integers are nonnegative integers. Consequently all common extensions in any closed disk of radius $R \geq 33$ can be generated from those in the closed disk of radius 33. Repeatedly apply both transformations (7.7), retaining new values only when their moduli are at most R. Reversing a decreasing chain proves completeness of this procedure. Zero and anchor coincidences must be retained until the generation is finished.

Lemma 7.6. *For every squarefree $d < 36$ and every compatible small pair $a, b \in \mathcal{O}_{\mathbb{Q}(\sqrt{-d})}$ with $ab + 1 \neq 0$, at most two members of a $D(1)$-tuple containing a, b have moduli in $[3, 10^{550}]$.*

Proof. We describe the complete finite calculation, including its arithmetic tests and search domains. Write $t = \sqrt{-d}$ and represent $(u+vt)/2$ by (u, v). Valid coordinates have matching parity when $d \equiv 3 \pmod 4$, and are both even otherwise. Their norm is $(u^2 + dv^2)/4$.

For an element $(u+vt)/2$, the following is a necessary and sufficient square test. Require

$$\frac{u^2+dv^2}{4}=L^2, \qquad U^2=2L+u, \qquad dV^2=2L-u, \qquad UV=|v|\,,$$

with nonnegative integers L, U, V, and require that (U, V) satisfy the coordinate parity rule. A square root is $(U+\epsilon Vt)/2$, where $\epsilon=1$ if $v\geq 0$ and $\epsilon=-1$ otherwise. Squaring this expression proves sufficiency. Conversely, the norm of a square gives L, and adding and subtracting its real-coordinate equations gives the displayed conditions. Thus the test and a choice of square root use only integer square roots, divisibility, and parity.

Enumerate the squarefree integers $1\leq d<36$. For each d, enumerate all nonzero valid coordinates with $u^2+dv^2<36$, form all distinct compatible pairs, and select the lexicographically least representative under simultaneous negation, conjugation, and interchange of the two entries. This gives 72 classes. Twenty-three are the pair $\{-1,1\}$, one in each field. One further class has product -1, namely

$$d=3, \qquad a=(-1-t)/2, \qquad b=(1-t)/2\,. \tag{7.8}$$

The remaining 48 representatives are listed in Table 2.

For each of these 48 pairs, enumerate every valid coordinate with $u^2+dv^2\leq 4\cdot 33^2$ and retain precisely its common extensions, including zero and any anchor coincidences. Close this finite initial set under both transformations (7.7), retaining a new value exactly when $u^2+dv^2\leq 4\cdot 10^{1100}$. Any root returned by the square test may be used, since both signs of the transformation are included. Continue until no new value is produced. The descent lemma proves that the resulting set is the complete closed radius-10^{550} set of common extensions.

Now discard zero, the two anchors, and entries of modulus below three. Join every pair of remaining vertices whose product plus one passes the square test. A triangle is detected by taking, for every edge $\{x,y\}$, the intersection of the two neighbor sets and testing whether it contains a third vertex. Table 2 gives the exact vertex and edge counts. The 48 graphs contain 36931 vertices and 36854 edges in total, and every triangle test is negative. Since any three compatible extensions in the disk would form a triangle, the assertion follows. □

Table 2: Nondegenerate small pairs and their complete extension graphs through modulus 10^{550}. A coordinate pair (u,v) represents $(u+v\sqrt{-d})/2$. Every graph has zero triangles.

d	First anchor	Second anchor	Vertices	Edges
1	$(-4,-4)$	$(-4,4)$	718	716
1	$(-4,-4)$	$(0,-4)$	414	413
1	$(-4,-4)$	$(4,0)$	827	825
1	$(-4,-2)$	$(0,-2)$	1192	1190
1	$(-4,-2)$	$(0,4)$	861	859
1	$(-4,-2)$	$(4,-2)$	1315	1312
1	$(-4,0)$	$(-2,-4)$	862	860
1	$(-4,0)$	$(2,0)$	1434	1432

Continued on the next page

d	First anchor	Second anchor	Vertices	Edges
1	$(-2,-4)$	$(2,-4)$	438	437
1	$(-2,-4)$	$(2,0)$	1192	1190
1	$(-2,-2)$	$(2,-2)$	718	717
1	$(0,-4)$	$(0,-2)$	718	717
2	$(-4,-2)$	$(0,-2)$	496	495
2	$(-4,0)$	$(2,-2)$	991	989
2	$(-2,-2)$	$(-2,2)$	961	959
2	$(-2,-2)$	$(2,-2)$	552	551
2	$(0,-4)$	$(0,4)$	718	716
3	$(-5,-1)$	$(-1,-1)$	524	523
3	$(-5,-1)$	$(3,1)$	892	890
3	$(-4,-2)$	$(0,-2)$	446	445
3	$(-4,-2)$	$(1,1)$	1048	1046
3	$(-4,0)$	$(4,0)$	962	960
3	$(-3,-1)$	$(-3,1)$	962	960
3	$(-3,-1)$	$(1,-3)$	892	890
3	$(-3,-1)$	$(1,-1)$	761	760
3	$(-3,-1)$	$(2,0)$	1522	1520
3	$(-2,-2)$	$(2,-2)$	481	480
3	$(-2,0)$	$(-1,-3)$	1048	1046
3	$(-1,-1)$	$(0,-2)$	761	760
3	$(0,-2)$	$(0,2)$	962	960
5	$(-2,-2)$	$(2,-2)$	410	409
6	$(-2,-2)$	$(2,-2)$	389	388
7	$(-5,-1)$	$(-5,1)$	718	716
7	$(-5,-1)$	$(-1,-1)$	457	456
7	$(-5,-1)$	$(0,2)$	737	735
7	$(-5,-1)$	$(2,0)$	1121	1119
7	$(-5,-1)$	$(5,-1)$	744	742
7	$(-3,-1)$	$(1,-1)$	561	560
7	$(-3,-1)$	$(3,1)$	914	912
7	$(-2,-2)$	$(-2,2)$	718	716
7	$(-2,-2)$	$(2,-2)$	372	371
11	$(-3,-1)$	$(1,-1)$	484	483
11	$(-1,-1)$	$(-1,1)$	962	960
15	$(-3,-1)$	$(1,-1)$	440	439
19	$(-3,-1)$	$(1,-1)$	411	410
23	$(-3,-1)$	$(-3,1)$	718	716
23	$(-3,-1)$	$(1,-1)$	389	388
31	$(-1,-1)$	$(-1,1)$	718	716
		Total	36931	36854

Lemma 7.7. *Let d be positive and squarefree, and put $t = \sqrt{-d}$. Every nonzero common extension of $\{-1, 1\}$ has one of the forms*

$$\begin{aligned} z &= \pm(dv^2 + 1), & u^2 - dv^2 &= 2, \\ z &= \pm 2uvt, & u^2 - dv^2 &= 1, \end{aligned}$$

where $u, v \in \mathbb{Z}$ and $v \neq 0$. In particular $|z| \geq d + 1$.

Proof. Choose integral roots $r^2 = z+1$ and $s^2 = 1-z$, and write $2r = U+Vt$, $2s = X+Yt$. The equation $r^2 + s^2 = 2$ gives

$$U^2 + X^2 - d(V^2 + Y^2) = 8, \qquad UV + XY = 0\,.$$

The real coordinate vector is nonzero, since otherwise the left side of the first equation would be nonpositive. If the imaginary coordinate vector vanished, both roots would be rational integers, forcing $r^2 = s^2 = 1$ and $z = 0$. Write $(U, X) = g(p, q)$ with $g > 0$ and $\gcd(p, q) = 1$. Orthogonality gives $(V, Y) = h(q, -p)$ for a nonzero integer h, so

$$(p^2 + q^2)(g^2 - dh^2) = 8\,.$$

Primitivity excludes $p^2 + q^2 = 4, 8$. Thus this sum is one or two. If it is one, one root is rational and the other purely imaginary. Integrality forces g, h even, including in the half-integral basis. Setting $u = g/2$, $v = h/2$ gives the first alternative.

If $p^2 + q^2 = 2$, then $p, q \in \{-1, 1\}$ and $g^2 - dh^2 = 4$. Outside the half-integral basis, g, h are even. In that basis they have matching parity. If both were odd, then $g^2 - dh^2 \equiv 2 \pmod 4$, contrary to its value four. Thus g, h are again even, and expansion gives the second alternative. The first has modulus at least $d + 1$, while in the second

$$|z|^2 = 4du^2v^2 \geq 4d(d + 1) > (d + 1)^2\,.$$

□

Lemma 7.8. *Suppose $\{-1, 1, c, d\}$ is a $D(1)$-quadruple, with $9 \leq C = |c| \leq D = |d|$ and $d \neq -c$. Then $D > C^3/2$.*

Proof. Choose $z \in \mathcal{O}_K$ with $z^2 = cd + 1$, and put $w = c + d$, $h = w/z^2$. Then $|h| \leq 2C/(C^2 - 1) \leq 9/40 < 1/4$. With the analytic square-root branches at zero, define $u = z\sqrt{1 + h}$, $v = z\sqrt{1 - h}$ and $L = u + v - 2z$. We have $u^2 = (c + 1)(d + 1)$ and $v^2 = (1 - c)(1 - d)$. Both products are squares in $\mathcal{O}_K$, so u, v, and hence L, lie in $\mathcal{O}_K$. If $L = 0$, squaring $u+v = 2z$ gives $uv = z^2$, and squaring again gives $w^2 = 0$. This contradicts $d \neq -c$, so $|L| \geq 1$. The even binomial coefficients have modulus at most $1/8$ after the constant term. Therefore

$$|L| \leq \frac{|z||h|^2}{4(1 - |h|^2)} < \frac{4}{15}\frac{|w|^2}{|z|^3} \leq \frac{16}{15}\left(\frac{81}{80}\right)^{3/2}\frac{\sqrt{D}}{C^{3/2}} < \sqrt{\frac{2D}{C^3}}.$$

Here $|w| \leq 2D$, $|z|^2 \geq (80/81)CD$, and $(16/15)^2(81/80)^3 < 2$. The conclusion follows. □

Lemma 7.9. *No $D(1)$-tuple in an imaginary quadratic integer ring contains $\{-1, 1\}$ and four further entries.*

Proof. We first restrict the axes and field parameters. A negative rational integer which is an integral square in $\mathbb{Q}(\sqrt{-d})$ is $-dm^2$ for an integer m. Indeed, the square of a rational linear combination of $1, t$ is rational only when one coordinate vanishes. Integrality of a purely imaginary root forces its coefficient of t to be an integer, also in the half-integral basis. A positive rational integer that is a square in $\mathcal{O}_K$ is therefore an ordinary integer square.

There are no nonzero common extensions when $d = 1$, since neither $u^2 - v^2 = 2$ nor $u^2 - v^2 = 1$ has a solution with $v \neq 0$. For $d > 1$, at most one extension can lie on each imaginary half-axis. Two with coefficients of the same sign would have product plus one equal to $1 - dkl < 0$, which is one modulo d and cannot be $-dm^2$.

Real and imaginary extensions cannot coexist in one tuple.

Suppose first that $\{-1, 1, a, -a, b\}$ is a $D(1)$-tuple with $|a|, |b| \geq 3$. The element $-b$ may also be adjoined. Indeed, $1 - b^2 = (1 - b)(1 + b)$ is a square, and its other compatibility conditions follow by changing signs. We may therefore interchange a, b and assume $A = |a| \leq B = |b|$. Choose roots $x^2 = 1 - b$, $y^2 = 1 + b$, $v^2 = 1 - ab$, $w^2 = 1 + ab$, and put

$$G = 2a^2b^2 - a^2 - 1, \qquad J = 2axyvw\,.$$

Expansion gives $(G - J)(G + J) = (a^2 - 1)^2$. Both factors are nonzero algebraic integers. Since $|G - J|, |G + J| \geq 1$, we have

$$|G - J| + |G + J| \leq |G - J||G + J| + 1.$$

Consequently

$$2|G| \leq (A^2 + 1)^2 + 1, \qquad |G| \geq 2A^4 - A^2 - 1\,.$$

These imply $3A^4 - 4A^2 - 4 \leq 0$, impossible for $A \geq 3$. Now if a real extension a and an imaginary extension b coexisted, conjugation would make $1 - ab$ a square, while $(1 - a)(1 + a)$ would make $1 - a^2$ a square. Adjoining $-a$ would give the forbidden five entries. Lemma 7.7 ensures that all these extensions have modulus at least three.

If d is even, a real extension has positive magnitude $u^2 - 1$ with u even. In $u^2 - dv^2 = 2$, an even v is impossible modulo four, and an odd v forces u even. Thus every positive magnitude is three modulo four. Two real extensions of the same sign would have a positive product plus one congruent to two modulo four. There are therefore at most two real extensions. If d is odd, an even v in the same Pell equation gives $u^2 \equiv 2$ or $6 \pmod 8$, which is impossible. An odd v gives $u^2 \equiv d + 2 \pmod 8$, forcing $d \equiv 7 \pmod 8$. We need only consider real extensions for odd squarefree $d \equiv 7 \pmod 8$.

The argument excluding $\{-1, 1, a, -a, b\}$ shows that the four real extensions have distinct absolute values. Write these as $S < T < C < D$. If $S \geq 10^4$, Lemma 7.8, applied three times, gives $D > S^{27}/2^{13} > S^{27}/10^4$. On the other hand, Corollary 4.3 with parameters

$$(h, r, u) = \left(\frac{163}{100}, -\frac{2}{5}, \frac{241}{100}\right)$$

gives, for $s = \log_{10} S$,

$$\log_{10} D < 7s + \frac{617}{25} + \frac{118401/2500}{s - 203/100}.$$

The parameters follow from $42^{100} < 10^{163}$, $(5/2)^5 < 10^2$ and $253^{100} < 10^{241}$. The difference from the lower bound $27s - 4$ is increasing for $s \geq 4$ and positive at 4, since $80 - 717/25 - 118401/4925 > 0$. Hence $S < 10^4$. Lemma 7.7 gives $d < S$, so it remains to consider squarefree $7 \leq d < 10^4$ with $d \equiv 7 \pmod 8$.

Put $R = 10^{550}$. We enumerate all real common extensions in the closed radius-R disk by ordinary continued fractions. By Lemma 7.7, their positive magnitudes are $u^2 - 1$, where

$$u^2 - dv^2 = 2, \qquad 1 \le u \le \lfloor \sqrt{R+1} \rfloor, \qquad v \ge 1.$$

Every such pair u, v is coprime, and

$$\left| \frac{u}{v} - \sqrt{d} \right| = \frac{2}{v^2(u/v + \sqrt{d})} < \frac{1}{2v^2}.$$

We recall a short proof of the criterion used here. If p/q is reduced and $|\alpha - p/q| < 1/(2q^2)$, choose consecutive convergents $p_n/q_n, p_{n+1}/q_{n+1}$ with $q_n \le q < q_{n+1}$. Their determinant is ± 1, and the errors $e_j = q_j\alpha - p_j$ alternate in sign. Indeed, the recurrence gives $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$ and, if $\alpha_{n+1} > 0$ is the next complete quotient,

$$\alpha = \frac{p_n \alpha_{n+1} + p_{n-1}}{q_n \alpha_{n+1} + q_{n-1}}.$$

These identities give the two assertions directly. If p/q is not a convergent, express $(p, q) = h(p_n, q_n) + k(p_{n+1}, q_{n+1})$ with integers h, k. The denominator bounds and reducedness force h, k to be nonzero and of opposite signs. Hence $|q\alpha - p| = |he_n + ke_{n+1}| \ge |e_n|$. But then

$$1 \le |pq_n - qp_n| \le (q_n + q)|q\alpha - p| \le 2q|q\alpha - p| < 1,$$

a contradiction. Applying this criterion with $\alpha = \sqrt{d}$ shows that every fraction u/v above is a convergent.

For completeness, the entire computation uses the following integer recurrences. For each admissible d, set $m_0 = 0$, $q_0 = 1$, and $a_0 = \lfloor \sqrt{d} \rfloor$, and iterate

$$m_{n+1} = q_n a_n - m_n, \qquad q_{n+1} = \frac{d - m_{n+1}^2}{q_n}, \qquad a_{n+1} = \left\lfloor \frac{a_0 + m_{n+1}}{q_{n+1}} \right\rfloor.$$

The convergent numerators and denominators satisfy

$$u_n = a_n u_{n-1} + u_{n-2}, \qquad v_n = a_n v_{n-1} + v_{n-2},$$

with $(u_{-2}, u_{-1}) = (0, 1)$ and $(v_{-2}, v_{-1}) = (1, 0)$. Stop at the first numerator exceeding $\lfloor \sqrt{R+1} \rfloor$, and retain precisely the convergents for which $v_n > 0$ and $u_n^2 - dv_n^2 = 2$. Add both vertices $\pm(u_n^2 - 1)$. The increasing numerators and Legendre's criterion prove that no extension in the disk is omitted.

Test every pair of distinct vertices by the rational square criterion at the start of this proof. As before, detect triangles by intersecting the neighbor sets at each edge. The complete computation gives the following counts.

Fields	Nonempty graphs	Vertices	Edges	Triangles
1016	409	19612	9806	0

Thus four compatible extensions would have their third modulus greater than 10^{550}, contrary to Lemma 7.4. □

Lemma 7.10. *A $D(1)$-tuple containing the pair in (7.8) has at most four entries.*

Proof. Put $t = \sqrt{-3}$, $a = (-1-t)/2$, and $b = (1-t)/2$. Then $a^3 = 1$, $b = -a^2 = -\overline{a}$, and $|a| = 1$. For a common extension z, choose roots $r^2 = az + 1$ and $s^2 = bz + 1$, and set $q = as$. Their equations give

$$r^2 + q^2 = 1 + a^2 = -a\,.$$

The integral transformation

$$T(r, q) = (2r + tq, -tr + 2q)$$

preserves this sum of squares and has integral inverse $T^{-1}(r, q) = (2r - tq, tr + 2q)$, since $4 + t^2 = 1$. Under the embedding $t = i\sqrt{3}$, the coordinates $u = r + iq$, $v = r - iq$ are multiplied by $\alpha = 2 + \sqrt{3}$ and α^{-1}, respectively. Their product is $-a$, so $|u||v| = 1$.

Choose an integer power of T for which $\alpha^{-1/2} \le |u| \le \alpha^{1/2}$. The transformed integral roots satisfy

$$|r|, |q| \le \frac{|u| + |v|}{2} \le \frac{\sqrt{\alpha} + 1/\sqrt{\alpha}}{2} = \frac{\sqrt{6}}{2}\,.$$

Their squared moduli are nonnegative integers, hence at most one. The six units in this ring are $\pm 1, \pm a, \pm a^2$, as follows by solving $U^2 + 3V^2 = 4$ with matching coordinate parity. Their squares are $1, a, a^2$. Since $-a$ is not one of those squares, neither root vanishes. The only two unit squares adding to $-a$ are 1 and a^2. Thus the transformed pair is $(\pm 1, \pm a)$ or $(\pm a, \pm 1)$, and satisfies

$$q = \epsilon a \overline{r}, \qquad \epsilon \in \{-1, 1\}\,.$$

This relation is preserved by T and its inverse. Indeed, if it holds, then $\epsilon a \overline{2r + tq} = 2q - tr$. Replacing t by $-t$ gives the identity for the inverse. It therefore holds for the original roots. As $q = as$, we have $s = \epsilon \overline{r}$, and the original equations give $bz = \overline{a}\,\overline{z}$. Since $b = -\overline{a}$, every common extension is purely imaginary.

A purely imaginary Eisenstein integer is kt with $k \in \mathbb{Z}$, by the coordinate parity rule. Two nonzero extensions on the same half-axis have product plus one equal to $1 - 3kl < 0$, with k, l positive integers. This is one modulo three and cannot be an integral square, since such a negative rational square would be $-3m^2$. At most one extension lies on each half-axis. □

Proposition 7.11. *A $D(1)$-tuple in any imaginary quadratic integer ring cannot contain two entries of modulus below three and four further entries of modulus at least three.*

Proof. Write $K = \mathbb{Q}(\sqrt{-d})$ with d positive and squarefree. If $d \ge 36$, every nonreal algebraic integer has modulus at least $\sqrt{d}/2 \ge 3$. The small entries are among $-2, -1, 1, 2$. Testing their six products plus one gives only the compatible pair $\{-1, 1\}$. The other values are $3, -1, -3$, and a negative rational square in this field is $-dm^2$. Lemma 7.9 excludes this case.

If $d < 36$, the complete enumeration in Lemma 7.6 gives 72 classes. The unit pairs are excluded by Lemma 7.9, and the exceptional product--1 pair by Lemma 7.10. For every other pair, Lemma 7.6 forces the third large modulus to exceed 10^{550}, contradicting Lemma 7.4. □

7.3 Three small entries

Lemma 7.12. *A $D(1)$-tuple in an imaginary quadratic integer ring has at most three entries of modulus less than 3. Up to simultaneous negation and conjugation, its small triples are precisely those in Table 3. The table lists all further nonzero common extensions of each triple with modulus at most $R = 10^{40}$.*

Proof. Write $K = \mathbb{Q}(t)$ with $t = \sqrt{-d}$, where d is positive and squarefree. Represent an algebraic integer by $(u + vt)/2$. The integers u, v are both even if $d \not\equiv 3 \pmod 4$, and have the same parity otherwise. Small nonzero elements satisfy $0 < u^2 + dv^2 < 36$.

If $d \geq 36$, all such elements are rational and belong to $\{-2, -1, 1, 2\}$. A square root of a product of two of them plus one has modulus at most $\sqrt{5} < 3$, so it too is rational. The only compatible pair is $\{-1, 1\}$. Thus a small triple requires $d < 36$.

For each squarefree $d < 36$, form the graph on the admissible pairs with $0 < u^2+dv^2 < 36$, joining distinct entries when their product plus one is a square. The required roots have norm at most 9, so the square test uses the squares of all admissible pairs (r, s) with $r^2 + ds^2 \leq 36$, including $(0, 0)$. Testing all triangles gives 24 triples in the ten orbits displayed in the table. Their orbit sizes are listed in its last column. No fourth small entry is adjacent to all three vertices of any triangle.

For the extension column we use the reversible descent of Lemma 7.5. Number the three entries in each displayed triple as $1, 2, 3$. In the successive ten rows choose the pairs

$$(1,3), (1,2), (1,2), (1,3), (2,3), (1,3), (1,3), (1,2), (1,3), (1,2).$$

Each chosen pair a, b satisfies $ab + 1 \neq 0$. Enumerate all its common extensions in the closed disk of radius 33, retaining zero and any anchor coincidences. This initial list is obtained by enumerating roots q with

$$N(q) \leq \left\lfloor \sqrt{N(a)33^2} \right\rfloor + 1, \qquad z = (q^2 - 1)/a,$$

then checking integrality, $N(z) \leq 33^2$, and the other square condition. Starting from this list, repeatedly append both values

$$a + b + (2ab + 1)z \pm 2\sqrt{ab + 1}\sqrt{az + 1}\sqrt{bz + 1}$$

when their moduli are at most R, until no new value is obtained. The descent lemma proves completeness, since reversing a strictly decreasing descent from any extension in the disk stays in that disk.

Every step uses exact integer arithmetic. To specify the root test throughout this procedure, an integral $z = (u+vt)/2$ is a square precisely when $N(z) = h^2$ for a nonnegative integer h, and there are nonnegative integers r, s satisfying

$$r^2 = 2h + u, \qquad ds^2 = 2h - u, \qquad rs = |v|,$$

with (r, s) admissible. The signs then construct an integral root $(r \pm st)/2$. This criterion includes zero.

The numbers of distinct pair extensions obtained, including zero and small entries, are respectively

$$61, 89, 64, 105, 71, 112, 112, 57, 83, 55.$$

Testing the third anchor condition and removing zero and the original three entries gives the extension column of the table. Thus the finite procedure proves all its assertions. □

d	j	Small triple	Extension	Orbit size
1	1	$\{-2-2t, -2+2t, 2\}$	60	2
1	2	$\{-2-t, -t, 2-t\}$	$15t$	2
1	3	$\{-2-t, 2t, 2-t\}$	$-40t$	2
1	4	$\{-2, -1-2t, 1\}$	$4+12t$	4
2	1	$\{-2, 1-t, 1+t\}$	-24	2
3	1	$\{(-3-t)/2, (-3+t)/2, 1\}$	8	2
3	2	$\{(-3-t)/2, (1-3t)/2, 1\}$	$-12+4t$	4
3	3	$\{(-1-t)/2, -t, (1-t)/2\}$	none	2
7	1	$\{(-5-t)/2, (-5+t)/2, 1\}$	24	2
7	2	$\{(-5-t)/2, t, (5-t)/2\}$	$-32t$	2

Table 3: Small triples and all their further nonzero extensions of modulus at most 10^{40}. Here $t = \sqrt{-d}$.

Lemma 7.13. *Fix three entries a, b, s of a $D(1)$-tuple with $1 \le A \le B \le S < 3$. Let v, e be two further entries with $1000 \le V = |v| \le E = |e|$. Their coherent form satisfies*

$$\begin{cases} 5V^2 < E < 405V^2, & L_3(v,e) = 0, \\ E > V^5/100, & L_3(v,e) \neq 0. \end{cases}$$

In particular, $E > 5V^2$ in either case.

Proof. First suppose $E \le 4V$. Put $x = 1/v$ and $y = 1/e$. The exact identity is

$$\frac{L_3}{\Delta_3\sqrt{ve}} = 1 - \sqrt{1+xy} + \frac{F_3}{abs}.$$

Here $|F_3| < V^{-3}$, since $(1000/999)^3 < 4$. Rationalization gives

$$|1 - \sqrt{1+xy}| > \frac{1}{3VE} \ge \frac{1}{12V^2} > \frac{1}{V^3} \ge \frac{|F_3|}{ABS}.$$

Thus $L_3 \neq 0$. But $V/2 < Z < 3V$, so Lemma 3.3 implies

$$|L_3| < \frac{270}{V} + \frac{27}{V^2} < 1,$$

contrary to integrality. Therefore $E > 4V$.

If $L_3 = 0$, Lemma 3.4 gives $5ABSV^2 < E < 15ABSV^2$. Since $1 \le ABS < 27$, this is the first alternative. Otherwise the same lemma gives

$$1 \le |L_3| < \frac{270}{V} + \frac{9Z}{4V^3}.$$

The first term is less than $1/2$. Hence $Z > 2V^3/9$ and

$$E > \frac{4V^5}{81} - \frac{1}{V} > \frac{V^5}{100}.$$

Finally $V^5/100 > 5V^2$ for $V \ge 1000$. □

Proposition 7.14. *A $D(1)$-tuple in an imaginary quadratic integer ring containing three entries of modulus less than 3 has at most five entries.*

Proof. Put $R = 10^{40}$ and fix the three small entries, with ordered moduli $1 \le A \le B \le S < 3$. Lemma 7.12 and Table 3 give at most one further entry of modulus at most R. If such an entry exists, its modulus is less than 85. These are complete closed-disk extension lists, obtained by the reversible descent of Lemma 7.5.

We record the approximation bounds needed below. For a further pair of ordered moduli $R \le V \le E$, the two small anchors give

$$E < 10^{60} V^7. \tag{7.9}$$

Indeed, in Corollary 4.3 we may take $h = 3$, $r = -2$, and $u = 4$. The squared small moduli are integers at most 8. The necessary comparisons are

$$42^2 \cdot 8^3 < 10^6, \qquad 253 \cdot 8 < 10^4, \qquad \frac{2}{15} > 10^{-2}.$$

With $v = \log_{10} V \ge 40$, that corollary consequently gives

$$\log_{10} E < 7v + 48 + \frac{224}{v-5} + \frac{2}{5} < 7v + 60.$$

The three small anchors also forbid the closed interval

$$10^{40} V^5 \le E \le 10^{-22} V^{11}. \tag{7.10}$$

Apply the normalized three-root estimate in Lemma 4.4. Its order-two forbidden interval contains (7.10). This follows from $A \ge 1$, $B, S < 3$, and the exact comparisons

$$\frac{512}{81}\, 608^4\, 3^{22} < 10^{40}, \qquad 2 \cdot 28^4 \left(\frac{10}{81}\right)^2 3^{14} < 10^{22}.$$

The estimate applies since $V \ge 10^{40} > 4S$ and $AV > 1000$.

If the extension of modulus at most R exists, use it as a fourth anchor of modulus $T < 85$. Lemma 7.3 additionally forbids

$$10^{38} V^3 \le E \le 10^{-29} V^7. \tag{7.11}$$

Its hypotheses hold because $V \ge R > 4T$ and $AV > 1000$. The displayed endpoints follow from

$$2 \cdot 10300^2 \cdot 3^{12} \cdot 85^{12} < 10^{38}, \qquad 2 \cdot 193^2 \cdot 3^8 \cdot 85^{10} < 10^{29}.$$

Suppose now that three further entries exist. First consider the case where all three moduli satisfy $R < U \le V \le W$. Apply Lemma 7.13 to their three pairs. If $L_3(u, v) \ne 0$, then

$$V > U^5/100, \qquad W > 5V^2 > U^{10}/2000.$$

If $L_3(u, v) = 0$ and $L_3(v, w) \ne 0$, then $V > 5U^2$ and $W > V^5/100 > U^{10}/2000$ as well. Either conclusion contradicts (7.9), since $U^3 > 10^{120} > 2000 \cdot 10^{60}$.

If both consecutive forms vanish, their quadratic bounds give

$$5U^2 < V < 405U^2, \qquad 125U^4 < W < 405^3 U^4.$$

As $125U^4 > 405U^2$, the form $L_3(u, w)$ cannot vanish. Its nonzero alternative gives $W > U^5/100$. Hence $U < 100 \cdot 405^3$, contrary to $U > R$.

It remains to consider the unique extension of modulus at most R, whose modulus is $T < 85$, together with two other entries of moduli $R < V \le W$. Lemma 3.5 applies because

$$V^2 > R^2 > 32 \cdot 3 \cdot 3 \cdot 85 \ge \frac{32BST}{A^3}.$$

At least one coherent form on (a, b, s) or (a, b, t) is therefore nonzero. Lemma 3.3 gives

$$1 < \frac{108375}{Z} + \frac{255Z}{V^3}, \qquad Z^2 = |vw + 1|.$$

Since $Z > V/2$ and $216750/V < 1/2$, we obtain $Z > V^3/510$. Consequently

$$W > \frac{V^5}{260100} - \frac{1}{V} > \frac{V^5}{10^6} > 10^{38}V^3.$$

The intervals (7.11) and (7.10) overlap because $V^2 > 10^{69}$. Their union forbids every value from $10^{38}V^3$ to $10^{-22}V^{11}$. But (7.9) gives $W < 10^{60}V^7 < 10^{-22}V^{11}$, since $V^4 > 10^{82}$. This is the final contradiction. □

Proof of Proposition 7.1. Choose six distinct entries, including two of the small entries. Lemma 7.12 shows that at most three are small. If exactly two are small, Proposition 7.11 excludes the four large entries. If three are small, Proposition 7.14 excludes the three remaining entries. □

8 Reduction of the first three entries

We now treat an ordered sextuple with

$$1 \le A < 3 \le B \le S \le T \le C \le D.$$

The preceding small-entry and all-large arguments reduce the proof to this case. All norms below are the field norms, so $N(z) = |z|^2$ is a nonnegative rational integer.

8.1 A common interval calculation

Put

$$\varepsilon = \frac{1}{4000}, \qquad h = \frac{4001}{3999}, \qquad \mu = \frac{51051}{400000}, \qquad \kappa = \frac{3535}{409600},$$
$$M_m = \frac{8}{9}\,608^2, \quad R_m = \frac{3920}{81}.$$

We record the interval calculation used twice below. The five finite families have parameters

	N_2	N_3	A_3	A_4	A_5
k	2	3	3	4	5
P	6	9	8	11	14
g	2	4	4	2	4
R_*	5/4	35/32	35/16	63/8	495/64.

Here R_* is the residual constant for the corresponding family in Table 1. Write their coefficient bounds as M_2, M_n, M_3, M_4, M_5, and their residual bounds as R_2, R_n, R_3, R_4, R_5. Write R_p for the adaptive residual. For the relabeled ordinary family on b, s, put

$$v = B/S, \qquad d_0 = |b - s|/S, \qquad \Phi = \max(d_0^6, vd_0^5), \qquad R_{bs} = \kappa\Phi.$$

The coefficient $M_{bs} = 4143$ is sufficient when no specialization at b, s is made.

The approximation inequalities proved above imply that D avoids every closed interval $[l/(1-\varepsilon), u/(1+\varepsilon)]$ in Table 4. Whenever Lemma 3.7 applies, it supplies all the size hypotheses used in this table. Both expressions in the maximum defining the lower endpoint of P must be checked. The ordinary family O_j in this table is available for every positive integer j.

Table 4: Forbidden-interval endpoints.

Family	l	u
Q	$4 \cdot 55^2 T^6 C^3$	$A^4 C^7/(9T^2)$
N_2	$4M_2^2 B^8 C^5$	$A^6 C^7/(4R_2^2 B^4)$
P	$4 \cdot 2400^2 \max(BS^3, B^3S^2)^2 C^5$	$A^6 C^9/(4R_p^2 S^4)$
A_3	$4M_3^2 B^{12} C^7$	$A^8 C^9/(4R_3^2 B^6)$
M	$4M_m^2 B^6 S^{16} C^5$	$A^6 C^{11}/(4R_m^2 B^4 S^{10})$
N_3	$4M_n^2 B^{14} C^7$	$A^8 C^{11}/(4R_n^2 B^6)$
$N_{3,bs}$	$4M_{bs}^2 S^{14} C^7$	$B^8 C^{11}/(4R_{bs}^2 S^6)$
A_4	$4M_4^2 B^{18} C^9$	$A^{10} C^{13}/(4R_4^2 B^8)$
A_5	$4M_5^2 B^{24} C^{11}$	$A^{12} C^{17}/(4R_5^2 B^{10})$
O_j	$36(42B^3C)^{2j} C/B^2$	$(2AC^2/(5B))^{2j}/C$

Suppose that

$$C > GA^7 B^5 S^3, \qquad G_*/R_{bs} \geq \rho, \tag{8.1}$$

where G_* is a pointwise coefficient valid in the first inequality at the actual triple a, b, s. The constant G and the lower bound ρ may be estimated separately. In Table 5, each displayed quotient is a sufficient lower bound for the preceding upper endpoint divided by the next lower endpoint, including the factor h. The first line instead compares the initial lower bound $D > A^2 C^7/(S^2 T^4)$ with the lower endpoint of Q.

Table 5: Sufficient overlap quotients, each required to exceed one.

Junction	Quotient	Additional substitution
START–Q	$1024A^{10}B^{10}(1-\varepsilon)/12100$	none
Q–N_2	$256A^8S^4/(36M_2^2 h)$	none
N_2–P, BS^3	$G^2A^{20}B^4/(16R_2^2\,2400^2 h)$	none
N_2–P, B^3S^2	$G^2A^{20}S^2/(16R_2^2\,2400^2 h)$	none
P–N_3	$G^2A^{20}/(16R_p^2 M_n^2 h)$	$S \geq B^2$
P–A_3	$G^2A^{20}/(16R_p^2 M_3^2 h)$	$S \geq B$
A_3–M	$G^4A^{36}/(16R_3^2 M_m^2 h)$	$S \leq B^2$
M–N_3	$G^4A^{34}B^2S^2/(16R_m^2 M_n^2 h)$	none
N_3–A_4	$G^2A^{22}B^4/(16R_n^2 M_4^2 h)$	$S \geq B^3$
N_3–$N_{3,bs}$	$G^4A^{36}B^8/(16R_n^2 M_{bs}^2 h)$	$S \leq B^3$
$N_{3,bs}$–A_4	$\rho^2A^{14}/(16M_4^2 h)$	pointwise bound
A_4–O_4	$G^4A^{38}B^2/(144R_4^2\,42^8 h)$	$S \geq B$
O_4–A_5	$G^4A^{36}/(4M_5^2(5/2)^8 h)$	$S \geq B$
A_5–O_5	$G^6A^{54}B^{10}/(144R_5^2\,42^{10} h)$	$S \geq B$
O_5–O_6	$G^6A^{52}B^4/(36 \cdot 42^{12}(5/2)^{10} h)$	$S \geq B$

Here the first two comparisons use $C > A^2T^5/S^2$ and $T > 2BS$. The others use (8.1). For example, before substitution, the N_3–A_4 quotient is

$$\frac{A^8C^2}{16R_n^2 M_4^2 B^{24} h}.$$

Substituting the gap and then $S \ge B^3$ gives its table entry. At the critical relabeled junction the raw quotient is $C^2/(16R_{bs}^2M_4^2B^{10}S^6h)$. Substitution gives the displayed ρ^2A^{14} bound. The remaining entries follow by the same divisions. All substitutions use a positive exponent for a lower bound or a negative exponent for an upper bound.

The chain begins START,Q, N_2, P. If $S \ge B^2$, continue directly to N_3. Otherwise use A_3, M, N_3. If $S \ge B^3$, continue to A_4. Otherwise use $N_{3,bs}, A_4$. Then use

$$A_4, O_4, A_5, O_5, O_6, O_7, \dots.$$

Both equality cases belong to a checked branch. With $H_0 = 42B^3C$ and $R_0 = 2AC^2/(5B)$,

$$R_0/H_0 = AC/(105B^4) > (G/105)A^8BS^3.$$

When the last quantity exceeds one, the O_5–O_6 comparison implies every later ordinary overlap. Since $H_0 > 1$, the lower endpoints tend to infinity. The initial lower bound and avoidance of all these overlapping closed intervals give a contradiction.

8.2 The five regions with an unbounded second entry

The following elementary phase estimate keeps the two distances in the gap and in the relabeled residual coupled. Put

$$\mathcal{Q} = \frac{|(a-b)(a-s)(b-s)|}{BS^2}.$$

If $B \ge qA$, $q \ge 1$, then

$$\mathcal{Q}^2\Phi \le F(q) := \max_{0\le x\le 2}(4-x^2)^4(1+q^{-2}+x/q)^2. \tag{8.2}$$

Indeed, writing $u = A/B$, $v = B/S$, $d_0 = |b-s|/S$, the identity $(a-b)(a-s) = a^2 - a(b+s) + bs$ and the parallelogram identity give

$$\mathcal{Q} \le d_0\{1 + u^2v + u\sqrt{2+2v^2-d_0^2}\} \le d_0\{1+q^{-2}+q^{-1}\sqrt{4-d_0^2}\}.$$

For $d_0 \ge v$, substitute $x = \sqrt{4-d_0^2}$. For $d_0 < v \le 1$, the same expression gives $\mathcal{Q}^2\Phi \le (1+1/q)^4 \le 16 < F(q)$. The unique maximizing interior point in (8.2) is the positive zero of

$$5x^2/q + 4(1+q^{-2})x - 4/q = 0. \tag{8.3}$$

The derivative changes sign there from positive to negative.

Proposition 8.1. *In the one-small-entry branch, every hypothetical sextuple satisfies*

$$N(a) = 1,\ B < 8, \qquad \text{or} \qquad N(a) = 2,\ B < 4.$$

Proof. Since $N(a)$ is an integer between one and eight, it is enough to exclude the five rows of Table 6. In each row $B \ge m$, $A^2 \ge n$, $A \le U$, and $q = m/U$. Lemma 2.5 gives $T > tABS$. Consequently

$$AT > tnm^2 \ge 54,$$

so Lemma 3.7 applies with $\lambda = 19/200$. In particular it supplies the size assumptions for all approximation estimates and the starting lower bound. Put $c_* = (9999/10000)(200/19)^2$,

$$Q_0 = 2(1+1/q)^2, \quad H = c_*t^5(1-\mu Q_0/m^2)^2.$$

The triangle inequalities give $\mathcal{Q} \le Q_0$. Weakening $\mu/(A^2B^2)$ to μ/m^2 therefore gives

$$C > \frac{H}{\mathcal{Q}^2} A^7 B^5 S^3.$$

For every row of Table 6, exact substitution shows $H/Q_0^2 > G$ and $H/(\kappa F_p) > \rho$, where $F(q) < F_p$. This establishes both bounds in (8.1).

Table 6: Five excluded regions and their gap bounds.

$N(a)$	m	n	U	t	q	F_p	G	ρ	R_p
1	8	1	1	27/8	8	269	7495	20687	7/50
2	4	2	3/2	3	8/3	368	1771	7973	7/20
3	3	3	7/4	11/4	12/7	540	598	3226	18/25
4	3	4	2	29/10	3/2	637	624	3507	47/50
5–8	3	5	17/6	3	18/17	1121	375	2217	201/100

The finite-family residuals are

$$R_k = R_* \frac{101}{200} \left(\frac{1 + 1/q}{2} \right)^{2k}.$$

The coefficient bounds and overlap margins are given in Table 7. If a root-column monomial after division by its anchor is $na^r b^s c^\ell$, then

$$r, s \ge 0, \quad 0 \le \ell \le k, \quad r + s = \ell + P - k - 2, \qquad \frac{|na^r b^s c^\ell|}{B^{P-2} C^k} \le |n| q^{-r} 1000^{\ell - k}.$$

Thus a strict upper bound for

$$\frac{101}{100g} \max_{\text{rows}} \sum_{\text{two root columns}} \sum_{r,s,\ell} |n| q^{-r} 1000^{\ell-k} \tag{8.4}$$

is a valid M_k. The explicit polynomial rows constructed earlier, applied to (8.4), give the displayed integers. The adaptive residual is the maximum of its four explicit positive polynomials at $x = 1/q$. Their maximum is strictly below the displayed R_p.

Table 7: Coefficient bounds for the five regions. The last column is a strict lower bound for every overlap quotient.

$N(a)$	M_2	M_n	M_3	M_4	M_5	margin
1	36	282	145	4605	38390	1.260
2	69	697	326	14771	171856	2.329
3	109	1306	663	34919	482131	1.166
4	130	1652	863	48019	701908	5.459
5–8	220	3553	1944	134712	2400382	1.321

Here and below terminating decimals denote the corresponding exact rational numbers. All the numerical verifications in these two tables can be performed as follows. For F_p, bisect $[0, 2]$ thirty times using the sign of (8.3), obtaining $[l, r]$. Then evaluate $(4 - l^2)^4(1 + q^{-2} + r/q)^2 < F_p$. Evaluate (8.4), the four residual polynomials, and the two bounds for G, ρ by rational arithmetic. In each row of Table 5, substitute $A^2 = n$ and $B = S = m$. All remaining exponents are nonnegative and every exponent of A is even, so this gives a lower bound throughout the region. The resulting fifteen quotients are all larger than the last column of Table 7. Finally $(G/105)n^4m^4 > 1$, proving the infinite-tail condition. The common interval calculation gives a contradiction in every region. □

8.3 The field and the finite first-pair list

Lemma 8.2. *Every common extension of* $\{1,3\}$ *in the full integer ring of an imaginary quadratic field is rational. A* $D(1)$*-tuple containing* $1,3$ *and a negative entry has at most four elements.*

Proof. First every common extension is rational. Otherwise choose roots $r=(e+f\sqrt{-\Delta})/2$ and $u=(g+h\sqrt{-\Delta})/2$ of $c+1$ and $3c+1$, respectively. Nonreality makes e,f,g,h nonzero. From $3r^2-u^2=2$, put

$$X=3e^2-g^2,\quad V=h^2-3f^2,\quad k=(eh-fg)/2.$$

If Δ is odd, then $e\equiv f$ and $g\equiv h \pmod 2$. If Δ is even, then e and g are even. In either case k is an integer. The imaginary and real parts give $3ef=gh$ and $X+\Delta V=8$. Multiplication gives $XV=12k^2$. Moreover $V=3f^2X/g^2$, so $X,V>0$. Also $k\neq 0$, since $k=0$ would give $g^2=3e^2$, which is impossible for nonzero integers. Hence

$$64=(X+\Delta V)^2\geq 4\Delta XV=48\Delta k^2\geq 144,$$

a contradiction.

Write a negative rational extension as $-x$, where x is a positive integer. Purely imaginary integral roots in $\mathbb{Q}(\sqrt{-d})$ are integer multiples of $\sqrt{-d}$, including in the half-integral full rings. Thus $x-1=du^2$, $3x-1=dv^2$, and $d\mid 2$. The case $d=1$ is impossible modulo three, so $d=2$ and x is odd. A positive extra entry p satisfies $p+1=r^2$, $3p+1=s^2$ with integer roots. Reduction modulo four forces r odd, hence $8\mid p$. Such a positive entry cannot coexist with $-x$, because the equation $px-1=2w^2$ has an odd left side. If negative entries occur, all extra entries are therefore negative odd integers. At most two can coexist, since among three odd absolute values two have the same residue modulo four and their product plus one is congruent to two. This includes the possible zero square value when an entry is -1. □

Lemma 8.3. *Under Proposition 8.1, either the first pair is* $(1,3)$ *up to simultaneous negation, or the field has absolute discriminant* $\Delta\leq 24$.

Proof. For a nonreal algebraic integer, $N(z)\geq\Delta/4$. If $\Delta\geq 25$, the condition $N(a)\leq 2$ therefore makes a rational. Simultaneous negation allows $a=1$, and then $B<8$. For $r^2=b+1$, $N(r)=|b+1|<9$. If $\Delta\geq 36$, this forces r rational. The only absolute negative fundamental discriminants between 25 and 35 are 31 and 35. At 35 a nonreal integer has integral norm at least nine. At 31 the only nonreal roots of norm below nine are $(\pm 1\pm\sqrt{-31})/2$. They give $N(r^2-1)=80$, contrary to $N(b)<64$. Thus r is rational in every case. Now $b=r^2-1$, $r\in\mathbb{Z}$, and $3\leq|b|<8$ force $b=3$, which is the stated alternative. □

We enumerate the remaining pairs in the fields allowed by Lemma 8.3. For squarefree

$$d\in\{1,2,3,5,6,7,11,15,19,23\},$$

write $\omega=(1+\sqrt{-d})/2$ when $d\equiv 3 \pmod 4$, and $\omega=\sqrt{-d}$ otherwise. Put $\omega^2=t\omega-u$, $\Delta=4u-t^2$. The exact arithmetic is

$$(x,y)(v,w)=(xv-uyw,xw+yv+tyw),\qquad N(x,y)=x^2+txy+uy^2. \tag{8.5}$$

All elements of norm at most L are given by

$$|y| \le \left\lfloor \sqrt{4L/\Delta} \right\rfloor, \quad |e| \le \left\lfloor \sqrt{4L - \Delta y^2} \right\rfloor, \quad e \equiv ty \pmod 2, \quad x = (e - ty)/2. \tag{8.6}$$

This follows from $4N(x, y) = (2x + ty)^2 + \Delta y^2$, and is both necessary and sufficient.

Enumerate every a of norm one or two and every r of norm at most eight. Form

$$b = \overline{a}(r^2 - 1)/N(a),$$

retaining it only when both coordinates are integers and

$$9 \le N(b) \le 63 \quad (N(a) = 1), \qquad 9 \le N(b) \le 15 \quad (N(a) = 2).$$

The root range is exhaustive because $N(r) = |ab + 1| \le AB + 1 < 9$ in the first case and $AB + 1 \le \sqrt{30} + 1 < 7$ in the second. Set aside $(a, b) = (1, 3), (-1, -3)$, which will be treated separately. Quotient only by simultaneous negation and conjugation of both coordinates. For an explicit representative rule, minimize lexicographically

$$\big(2a_x + ta_y < 0,\ a_y < 0,\ 2b_x + tb_y < 0,\ b_y < 0,\ a_x, a_y, b_x, b_y\big)$$

in each orbit, with false preceding true. The result is exactly the forty pairs in Table 8. The number of orbits and the number before these symmetries, respectively, are

d	1	2	3	5	6	7	11	15	19	23
orbits	9	3	15	2	2	2	2	2	2	1
pairs	30	10	58	6	6	8	8	8	8	4.

Their totals are forty and 146. The symmetries preserve every square condition. No arbitrary unit multiplication is used.

Table 8: The forty first-pair representatives. Coordinates mean $x + y\omega_d$.

i	d	a	b	$N(a)$	$N(b)$
0	1	$(0, 1)$	$(0, -3)$	1	9
1	1	$(0, 1)$	$(0, 5)$	1	25
2	1	$(0, 1)$	$(4, -2)$	1	20
3	1	$(0, 1)$	$(4, 4)$	1	32
4	1	$(1, 0)$	$(-5, 0)$	1	25
5	1	$(1, 0)$	$(-4, 4)$	1	32
6	1	$(1, 0)$	$(2, 4)$	1	20
7	2	$(1, 0)$	$(-3, 0)$	1	9
8	2	$(1, 0)$	$(-2, 2)$	1	12
9	2	$(1, 0)$	$(1, 4)$	1	33
10	3	$(0, 1)$	$(-9, 6)$	1	63
11	3	$(0, 1)$	$(-6, 9)$	1	63
12	3	$(0, 1)$	$(-5, 1)$	1	21
13	3	$(0, 1)$	$(-4, -4)$	1	48
14	3	$(0, 1)$	$(-4, 4)$	1	16
15	3	$(0, 1)$	$(-1, 5)$	1	21
16	3	$(0, 1)$	$(2, -7)$	1	39
17	3	$(0, 1)$	$(3, -3)$	1	9

i	d	a	b	$N(a)$	$N(b)$
18	3	$(0,1)$	$(4,4)$	1	48
19	3	$(0,1)$	$(7,-2)$	1	39
20	3	$(1,0)$	$(-9,3)$	1	63
21	3	$(1,0)$	$(-5,4)$	1	21
22	3	$(1,0)$	$(-4,0)$	1	16
23	3	$(1,0)$	$(-4,8)$	1	48
24	3	$(1,0)$	$(2,5)$	1	39
25	5	$(1,0)$	$(-6,0)$	1	36
26	5	$(1,0)$	$(-5,2)$	1	45
27	6	$(1,0)$	$(-7,0)$	1	49
28	6	$(1,0)$	$(-6,2)$	1	60
29	7	$(1,0)$	$(-2,3)$	1	16
30	7	$(1,0)$	$(1,5)$	1	56
31	11	$(1,0)$	$(-4,1)$	1	15
32	11	$(1,0)$	$(-3,3)$	1	27
33	15	$(1,0)$	$(-5,1)$	1	24
34	15	$(1,0)$	$(-4,3)$	1	40
35	19	$(1,0)$	$(-6,1)$	1	35
36	19	$(1,0)$	$(-5,3)$	1	55
37	23	$(1,0)$	$(-7,1)$	1	48
38	1	$(1,1)$	$(-1,-3)$	2	10
39	1	$(1,1)$	$(3,1)$	2	10

Table 9: Coefficient bounds for the forty fixed pairs.

i	M_2	M_n	M_3	M_4	M_5
0	62	608	280	12222	136530
1	17	77	44	808	4011
2	37	281	147	4610	38256
3	21	110	64	1402	8160
4	44	379	180	6340	61370
5	37	293	150	4791	40361
6	23	129	77	1798	11204
7	62	608	280	12222	136530
8	45	377	183	6284	60438
9	25	158	95	2471	16976
10	24	156	94	2424	16601
11	21	119	70	1599	9691
12	34	252	136	4117	32986
13	37	286	147	4661	38983
14	22	121	72	1618	9751
15	17	73	41	741	3580
16	39	318	159	5209	45613
17	48	401	193	6835	66703
18	20	106	62	1336	7678
19	24	152	91	2326	15710
20	36	276	143	4502	37275
21	40	323	162	5308	46658
22	51	456	205	8194	83910
23	26	173	103	2780	19826

i	M_2	M_n	M_3	M_4	M_5
24	21	113	66	1460	8607
25	41	333	164	5461	49340
26	35	273	142	4450	36676
27	38	303	153	4954	42245
28	34	259	137	4205	34076
29	32	224	127	3666	28238
30	23	137	81	1973	12709
31	50	440	201	7791	78839
32	32	230	127	3748	29208
33	44	367	176	6049	57935
34	32	229	126	3718	28965
35	40	324	161	5305	46974
36	31	226	124	3660	28401
37	37	296	151	4834	40902
38	77	803	387	17946	216405
39	19	38	41	370	985

For the rational pair $(a, b) = (1, 3)$, the same coefficient calculation gives

$$(M_2, M_n, M_3, M_4, M_5) = (12, 41, 25, 300, 1100).$$

These constants are independent of the imaginary quadratic field.

8.4 A common bound $S < 15$

Proposition 8.4. *No ordered sextuple whose first pair is* $(1, 3)$ *or one of the pairs in Table 8 has* $S \geq 15$.

Proof. Suppose that $S \geq 15$. Lemma 2.2 gives $T > 2BS \geq 90$. We first prove the common estimate

$$T > \tau ABS, \qquad \tau = \frac{59}{20}. \tag{8.7}$$

For the edges $12, 13, 14, 23, 24, 34$ of (a, b, s, t), use the bounds

$$r = \left(\frac{1}{3}, \frac{1}{15}, \frac{1}{90}, \frac{1}{45}, \frac{1}{270}, \frac{1}{1350}\right), \qquad k = \frac{1}{10000}(5506, 5087, 5014, 5029, 5005, 5001).$$

For every corresponding pair r_e, k_e, we have $1/2 < k_e < 1$ and $r_e k_e^2 < 2k_e - 1$. Hence, for $z_e = 1/(a_i a_j)$ and $\delta_e = \sqrt{1 + z_e} - 1$ on the branch with positive real part,

$$|\delta_e| \leq k_e |z_e|, \qquad |\delta_e - z_e/2| \leq (k_e - 1/2)|z_e|.$$

The first inequality follows by rationalization and the second from $\delta - z/2 = -\delta^2/2$.

Let M be the integral matrix of order four defined in the proof of Lemma 2.5, with $(a_1, a_2, a_3, a_4) = (a, b, s, t)$. It is nonsingular by the even-dimensional congruence used there. Apply the determinant expansion (2.7)–(2.8). The coefficient bounding the term $T/(ABS)$ is $\alpha = 2k_{12}k_{13}k_{23}$. For complementary pairs $\{i, j\}$ and $\{k, l\}$, the coefficient bounding $1/(A_i A_j)$ is

$$c_{ij} = 2\{k_{ij}(k_{ik}k_{jl} + k_{il}k_{jk}) + k_{ij}^2 k_{kl}\} - \frac{1}{2}.$$

Indeed, retain the signed leading cubic in (2.8) and bound each cubic remainder by the product of its three k's minus $1/8$, counting repeated edges with their multiplicities. Put

$$Q_4 = (k_{12}k_{34} + k_{13}k_{24} + k_{14}k_{23})^2, \qquad W = \frac{1}{4050}.$$

Since $A < 3$, $B \geq 3$, $S \geq 15$ and $T > 2BS$, the three remaining cubic terms $A/(BST)$, $B/(AST)$ and $S/(ABT)$ are at most $3W$, $1/450$ and $1/18$, respectively. The quartic factor $1/(ABST)$ is at most W. Consequently

$$1 \leq |\det M| \leq \alpha \frac{T}{ABS} + \beta,$$

where

$$\beta = 6k_{23}k_{24}k_{34}W + \frac{2k_{13}k_{14}k_{34}}{450} + \frac{2k_{12}k_{14}k_{24}}{18} + \sum_{i<j} c_{ij}r_{ij} + Q_4 W.$$

Substitution of the six rational k_e gives $\alpha < 141/500$ and $\beta < 21/125$. If $T \leq (59/20)ABS$, these inequalities would give

$$1 < \frac{141}{500}\frac{59}{20} + \frac{21}{125} = \frac{9999}{10000},$$

a contradiction. This proves (8.7). Since $AT > 90$, Lemma 3.7 applies with $\lambda = 1/10$.

Write $n_a = N(a)$, $n_b = N(b)$ and $n_{ab} = N(a-b)$. For a nonnegative rational x, define

$$\lfloor \sqrt{x} \rfloor_j = 10^{-j} \lfloor 10^j \sqrt{x} \rfloor, \qquad \lceil \sqrt{x} \rceil_j = 10^{-j} \lceil 10^j \sqrt{x} \rceil.$$

These bounds use only integer square roots and comparisons. Put

$$A_1 = \lceil \sqrt{n_a} \rceil_8, \quad B_0 = \lfloor \sqrt{n_b} \rfloor_8, \quad B_1 = \lceil \sqrt{n_b} \rceil_8, \quad c_0 = \frac{9999}{100},$$

and define

$$\begin{aligned} d_1 &= \lceil \sqrt{n_{ab}/n_b} \rceil_8, & Q_0 &= d_1(1 + A_1/15)(1 + B_1/15), \\ K &= c_0 n_a (1/Q_0 - \mu/n_b)^2, & G &= K\tau^5, \\ H &= c_0 n_a \tau^5 (1 - \mu Q_0/n_b)^2, & F_0 &= (n_{ab}/n_b)(1 + A_1/15)^2(1 + B_1/15)^8, \\ \rho &= H/(\kappa F_0). \end{aligned}$$

For all forty-one pairs, $1/Q_0 > \mu/n_b$ and $K > 7/4$. The retained gap, after replacing $\mu/(A^2B^2)$ by μ/B^2, gives

$$C > KT^5/S^2 > \frac{7}{4}T^5/S^2, \qquad C > GB^5S^3.$$

Moreover, $\mathcal{Q} \leq Q_0$ and $\mathcal{Q}^2\Phi \leq F_0$. For the second inequality, use $|a-s|/S \leq 1 + A/S$, $|b-s|/S \leq 1 + B/S$ and $\Phi \leq (1 + B/S)^6$. Thus the pointwise coefficient of B^5S^3, divided by R_{bs}, is at least ρ. All other positive powers of A in the interval endpoints may be replaced by one.

We evaluate the finite-family coefficients at a, b before taking absolute values. If the two root columns after division by their anchors are $\sum_{\ell=0}^{k} p_{j,\ell}(a,b)c^\ell$, put $C_0 = \max(1000, 200B)$. Any integer strictly larger than

$$\frac{101}{100g} \max_{\text{rows}} \sum_{j=1}^{2} \sum_{\ell=0}^{k} \frac{|p_{j,\ell}(a,b)|}{B^{P-2} C_0^{k-\ell}}$$

is a valid coefficient bound, since $C \ge C_0$. To reproduce the displayed integers, write $n_p = N(p_{j,\ell}(a,b))$ and $r = k - \ell$, and replace each summand before $101/(100g)$ by

$$\begin{cases} \lceil \sqrt{n_p / n_b^{P-2}} \rceil_{10} / 1000^r, & n_b \le 25, \\ \lceil \sqrt{n_p / n_b^{P-2+r}} \rceil_{10} / 200^r, & n_b > 25. \end{cases}$$

Take the integer part of the largest rational row sum, plus one. Every polynomial evaluation uses (8.5).

For every pair, $|a-b| \ge B - A > A$. This also holds for the two norm-two pairs, since $B \ge 3 > 2\sqrt{2}$. Hence the residual constants are

$$R_k = R_* \frac{101}{200} \left(\frac{n_{ab}}{4 n_b} \right)^k .$$

For the adaptive residual use $q = B_0/A_1 > 2$ and $x = 1/q$. Its largest residual bound is

$$R_p = \frac{15 + 80x + 150x^2 + 280x^3 + 35x^4}{256} + \frac{21(885 + 1144x + 435x^2 + 2634x^3 + 170x^4)}{1024 \cdot 999}.$$

The comparison with the other three residual polynomials follows directly from $0 \le x \le 1$ and $x^5 \le x^4 \le x^3$.

Use Table 5 with every A replaced by one. The first two numerators improve from $1024, 256$ to $9604, 784$, respectively, because $C > (7/4)T^5/S^2$. Substitute $B^2 = n_b$ and $S^2 = 225$ in the remaining nonnegative powers. For each of the forty-one pairs, all fifteen rational quotients are strictly larger than $59/50$. These checks use only the displayed formulas and family polynomials. The infinite-tail condition holds as well, since the same constants give

$$\frac{R_0}{H_0} \ge \frac{C}{105 B^4} > \frac{GBS^3}{105} \ge \frac{GB_0\, 15^3}{105} > 1.$$

The common interval argument gives the contradiction. □

8.5 The remaining third entries

The remaining third entries are found by a bounded enumeration. For each pair of Table 8, enumerate s by (8.6) with $L = 224$, retain $N(s) \ge N(b)$ and $s \ne a, b$, and test $as + 1$ and $bs + 1$ for being squares. The norm cutoff is exact because Proposition 8.4 gives $S < 15$ and the norm is integral. An exact full-ring square test, including zero, is as follows. For $z = (z_x, z_y)$, put $p = 2z_x + t z_y$, $q = z_y$. Require $N(z) = n^2$ with $n \ge 0$ integral, then require integers $e \ge 0, f$ satisfying

$$e^2 = 2n + p, \quad \Delta f^2 = 2n - p, \quad ef = q, \quad e \equiv tf \pmod 2.$$

If they exist, $((e - tf)/2, f)$ is an integral root, and conversely every root gives these equations after changing its sign if necessary. When $e = 0$, either sign of f gives the same square. A final multiplication verifies every retained witness.

This calculation gives 57 triples. One is $(1, -3, 3)$, already excluded by Lemma 8.2. There remain exactly 56 triples, retaining all equal-modulus cases and without any additional symmetry on the third entry. They are listed, together with their cutoffs N_0, in Table 11. It remains to exclude these fifty-six triples and the pair $(1, 3)$.

9 Completion for one small entry

Proposition 8.4 gives $S < 15$. For the forty pairs in Table 8, it leaves fifty-six triples. By Lemma 8.2, any sextuple containing the additional pair $(1, 3)$ consists of positive rational integers. The conditions $3 < s < 15$, $s + 1$ square and $3s + 1$ square give $s = 8$. For each of these fifty-six triples and for $(1, 3, 8)$, the forbidden intervals give a bound $N(t) < N_0$. We then contradict this bound using the growth of an irregular triple, with a separate square calculation for the last two cases.

9.1 A common fourth-entry cutoff

Fix one of these triples (a, b, s) and put

$$a_0 = N(a), \quad b_0 = N(b), \quad s_0 = N(s), \qquad d_{xy} = N(x - y), \quad L = 10^{10}.$$

For a nonnegative rational number v define rational enclosures

$$[v]_- = \frac{\lfloor L\sqrt{v} \rfloor}{L}, \qquad [v]_+ = \frac{\lceil L\sqrt{v} \rceil}{L}.$$

These definitions require only integer square comparisons. The lower integer n is characterized by $n^2 \le L^2 v < (n + 1)^2$. No floating-point comparison is used below. Define

$$q_1 = \left[\frac{d_{ab} d_{as} d_{bs}}{b_0 s_0^2} \right]_+, \qquad K = \frac{9999}{100}\, a_0 \left(\frac{1}{q_1} - \frac{\mu}{b_0} \right)^2,$$
$$\phi_1 = \left[\max\left\{ \frac{d_{bs}^6}{s_0^6}, \frac{b_0 d_{bs}^5}{s_0^6} \right\} \right]_+, \qquad R_{bs} = \frac{3535}{409600}\phi_1, \qquad \mu = \frac{51051}{400000}.$$

In every row $1/q_1 > \mu/b_0$ and $K > 1$. For $a_0 = 1$, Lemma 2.2 gives $T > 2BS = 2ABS$. For $a_0 = 2$ we will impose $N(t) \ge 4a_0 b_0 s_0 + 1$. Thus, whenever $N(t) \ge 1024$ and this additional condition holds, the row $H = 32$, $\lambda = 1/10$ of Lemma 3.7 implies

$$C > K\frac{T^5}{S^2}, \qquad D > \frac{C^7}{S^2 T^4}. \tag{9.1}$$

Indeed, $\mathcal{Q} \le q_1$ and $\mu/(a_0 b_0) \le \mu/b_0$, so the coefficient of T^5/S^2 in Lemma 3.7 is at least K. The other enclosure satisfies $\Phi \le \phi_1$, so the displayed R_{bs} is a valid residual constant for $N_{3,bs}$. The same lemma gives $C > \max(1000, 200S, 4T)$ and $D > 1000C$, which supply all approximation hypotheses in this section.

Use the constants M_2, M_n, M_3, M_4, M_5 in Table 9. For $(a, b) = (1, 3)$, take $12, 41, 25, 300, 1100$, respectively. Their normalizations are those of N_2, N_3, A_3, A_4, A_5. For $N_{3,bs}$ evaluate the same N_3 polynomials at b, s and use (5.13) with $C_0 = \max(1000, 200S)$. More explicitly, in the notation of that formula, replace every summand $|p_{ij\ell}| C_0^{\ell-3}/S^7$ by

$$\begin{cases} [N(p_{ij\ell})/s_0^7]_+/1000^{3-\ell}, & s_0 \le 25, \\ [N(p_{ij\ell})/s_0^{10-\ell}]_+/200^{3-\ell}, & s_0 > 25. \end{cases}$$

Multiply the largest row sum by $101/400$ and take its floor plus one. Denote this integer by M_{bs}. This is the integer printed below. Evaluation precedes taking norms, so cancellation between monomials of the same c-degree is preserved.

Since $|a-b| \ge B - A > A$ for all these pairs, the distance-sensitive residual constants in the first five families are

$$R_k = R_* \frac{101}{200} \left(\frac{d_{ab}}{4b_0} \right)^k, \qquad (R_*, k) = (5/4, 2), (35/32, 3), (35/16, 3), (63/8, 4), (495/64, 5).$$

They will be denoted by R_2, R_n, R_3, R_4, R_5, respectively. For P take R_p from (5.16) with $q = [b_0/a_0]_-$, and take its coefficient bound 2400. Finally put $M_m = (8/9)608^2$ and $R_m = 3920/81$.

For completeness, all raw interval endpoints used here are displayed in Table 10. A row (l, u) excludes $[l/(1-\varepsilon), u/(1+\varepsilon)]$, with $\varepsilon = 1/4000$. Positive powers of $A \ge 1$ have been discarded only from upper endpoints, so the same table applies to both possible values of a_0.

row	l	u
Q	$4 \cdot 55^2 T^6 C^3$	$C^7/(9T^2)$
N_2	$4M_2^2 B^8 C^5$	$C^7/(4R_2^2 B^4)$
P	$4 \cdot 2400^2 \max(BS^3, B^3S^2)^2 C^5$	$C^9/(4R_p^2 S^4)$
A_3	$4M_3^2 B^{12} C^7$	$C^9/(4R_3^2 B^6)$
M	$4M_m^2 B^6 S^{16} C^5$	$C^{11}/(4R_m^2 B^4 S^{10})$
N_3	$4M_n^2 B^{14} C^7$	$C^{11}/(4R_n^2 B^6)$
$N_{3,bs}$	$4M_{bs}^2 S^{14} C^7$	$B^8 C^{11}/(4R_{bs}^2 S^6)$
A_4	$4M_4^2 B^{18} C^9$	$C^{13}/(4R_4^2 B^8)$
A_5	$4M_5^2 B^{24} C^{11}$	$C^{17}/(4R_5^2 B^{10})$
O_j	$36(42B^3C)^{2j} C/B^2$	$(2C^2/(5B))^{2j}/C$

Table 10: Intervals for the fifty-six listed triples and $(1, 3, 8)$.

Put $h = 4001/3999$ and $x = N(t)$. The next table gives a strict lower bound $c_e x^{p_e}$ for each ratio of endpoints, using only (9.1). For the first row the ratio is $(1-\varepsilon)C^7/(S^2T^4 l_Q)$. For every other row it is $u_{\text{left}}/(h l_{\text{right}})$. Both terms in the maximum defining P are checked separately.

junction	$c_e x^{p_e}$
start–Q	$(1-\varepsilon)K^4x^5/(12100 s_0^5)$
Q–N_2	$K^2x^4/(36M_2^2 b_0^4 s_0^2 h)$
N_2–P, first term	$K^2x^5/(16R_2^2 \cdot 2400^2 b_0^3 s_0^5 h)$
N_2–P, second term	$K^2x^5/(16R_2^2 \cdot 2400^2 b_0^5 s_0^4 h)$
P–A_3	$K^2x^5/(16R_p^2 M_3^2 b_0^6 s_0^4 h)$
A_3–M	$K^4x^{10}/(16R_3^2 M_m^2 b_0^6 s_0^{12} h)$
M–N_3	$K^4x^{10}/(16R_m^2 M_n^2 b_0^9 s_0^9 h)$
N_3–$N_{3,bs}$	$K^4x^{10}/(16R_n^2 M_{bs}^2 b_0^3 s_0^{11} h)$
$N_{3,bs}$–A_4	$K^2x^5/(16R_{bs}^2 M_4^2 b_0^5 s_0^5 h)$
A_4–O_4	$K^4x^{10}/(144R_4^2 \cdot 42^8 b_0^{15} s_0^4 h)$
O_4–A_5	$(2/5)^8 K^4x^{10}/(4M_5^2 b_0^{16} s_0^4 h)$
A_5–O_5	$K^6x^{15}/(144R_5^2 \cdot 42^{10} b_0^{19} s_0^6 h)$
O_5–O_6	$(2/5)^{10} K^6x^{15}/(36 \cdot 42^{12} b_0^{22} s_0^6 h)$

For each row let n_e be the least positive integer with $c_e n_e^{p_e} > 1$. Doubling gives an upper bound, after which integer binary search determines n_e. The final checks are $c_e n_e^{p_e} > 1$ and $c_e (n_e - 1)^{p_e} \le 1$. Let n_∞ be defined in the same way by

$$\frac{K^2 x^5}{105^2 b_0^4 s_0^2} > 1,$$

and define

$$N_0 = \max\Big(1024, n_\infty, \{n_e\}_e, \eta\Big), \qquad \eta = \begin{cases} 1, & a_0 = 1, \\ 4a_0b_0s_0 + 1, & a_0 = 2. \end{cases}$$

All numbers and exponents in this prescription are rational or integral. The values in Table 11 result directly from these formulas.

If $N(t) \geq N_0$, every overlap is strict. The starting lower bound and successive forbidden intervals force D beyond the upper endpoint of each of

$$Q, N_2, P, A_3, M, N_3, N_{3,bs}, A_4, O_4, A_5, O_5, O_6, \ldots.$$

The squared ratio of the ordinary bases is greater than $K^2x^5/(105^2b_0^4s_0^2) > 1$. Hence the checked overlap O_5–O_6 implies every later consecutive overlap. The lower endpoints tend to infinity. This contradicts the fixed value of D, and proves $N(t) < N_0$ for each of these triples.

For $(a, b, s) = (1, 3, 8)$ the formulas give

$$q_1 = \frac{1822916667}{5000000000}, \qquad K > 744, \qquad M_{bs} = 34, \qquad N_0 = 1024.$$

For clarity, the thirteen integers n_e, in the order of the overlap table, are

$$3, 23, 96, 101, 108, 179, 131, 9, 34, 76, 123, 29, 438,$$

and $n_\infty = 14$. Thus $t < 32$, whereas Lemma 2.2 gives $t > 2 \cdot 3 \cdot 8 = 48$. This contradiction excludes the rational pair.

9.2 The finite complement

The fifty-six triples in Table 11 were enumerated at the end of Section 8. We now exclude $N(t) < N_0$. Lemma 2.2 gives $T > 2BS$. In thirty-eight rows, $4b_0s_0 \geq N_0$, so this already contradicts $N(t) < N_0$.

For the other rows, put

$$q = b + s, \qquad h = 1 + 2bs, \qquad \delta = (b - s)^2 - 4,$$

and define

$$f(X) = X^2 - 2(|h| - |q|)X + |\delta| + 2|q| + 1.$$

The triple (b, s, t) is irregular by Lemma 2.2, so (2.1) gives $f(T) \geq 0$. This quadratic is convex. For $X \geq 0$, the bounds $|h| \geq 2BS - 1$, $|q| \leq B + S$ and $|\delta| \leq (B + S)^2 + 4$ give

$$f(X) \leq (B + S + X + 1)^2 + 4 - 4BSX.$$

The endpoint calculation in the proof of Lemma 2.1 therefore gives $f(2BS) < 0$.

For sixteen of the remaining rows, substitution of the displayed entries and N_0 gives $f(\sqrt{N_0}) < -100$. These are exact checks using the rational enclosures defined above. Explicitly, $[N(h)]_- - [N(q)]_+ > 0$, and

$$\begin{aligned} f(\sqrt{N_0}) \leq N_0 &- 2\Big([N(h)]_- - [N(q)]_+\Big)[N_0]_- \\ &+ [N(\delta)]_+ + 2[N(q)]_+ + 1 < -100. \end{aligned}$$

Convexity excludes $2BS < T < \sqrt{N_0}$ in each of these rows.

The two remaining triples are $(a, b, s) = (1, -3, -2 \pm 2\sqrt{-2})$, both with $N_0 = 1024$. Here $N(q) = 33$, $N(h) = 457$ and $N(\delta) = 153$, so

$$f(30) = 901 - 60\sqrt{457} + 62\sqrt{33} + \sqrt{153} < -5.$$

Use $\sqrt{457} > 213/10$, $\sqrt{33} < 29/5$ and $\sqrt{153} < 62/5$. Convexity gives $T > 30$. Write $t + 1 = (u + v\sqrt{-2})^2$ with $u, v \in \mathbb{Z}$. If $30 < T < 32$, then

$$29 < |t + 1| = u^2 + 2v^2 < 33.$$

The values 30 and 31 are impossible modulo 16. For $u^2 + 2v^2 = 32$, reduction modulo 16 forces $4 \mid u, v$, and then $u = 0$, $v = \pm 4$. This gives $t = -33$, contrary to $T < 32$. Thus $N(t) < N_0$ is impossible in the last two rows as well.

Table 11: All remaining triples and the common cutoff procedure. Pair indices refer to Table 8. Coordinates mean $u + v\omega$.

Pair	u	v	$N(s)$	M_{bs}	N_0
0	-4	-2	20	285	1024
0	4	-2	20	285	1024
0	0	5	25	1368	1934
1	0	10	100	42	2311
2	-4	4	32	2238	4770
2	8	1	65	72	2359
3	6	9	117	44	3896
4	-4	-4	32	396	3299
4	-4	4	32	396	3299
5	-1	8	65	172	4857
6	7	6	85	29	2018
7	-2	-2	12	474	1024
7	-2	2	12	474	1024
8	1	4	33	133	1024
9	6	6	108	56	4775
10	-13	13	169	77	13059
11	-8	16	192	60	12244
12	-9	6	63	74	2301
13	-10	1	91	180	9744
14	-6	9	63	35	1195
15	-1	10	91	37	1760
16	-4	-4	48	430	7354
16	8	-8	64	271	7441
17	7	-2	39	78	1024
18	6	9	171	48	7943
19	11	1	133	48	6361
20	-6	-3	63	473	14404
20	-10	9	91	272	14608
21	-4	8	48	155	2461
22	-5	4	21	422	1622

Pair	u	v	$N(s)$	M_{bs}	N_0
22	-1	-4	21	422	1622
23	4	-8	48	4138	21270
23	-1	12	133	78	8968
24	7	7	147	41	5928
25	-5	-2	45	381	5913
25	-5	2	45	381	5913
26	-2	4	84	186	8435
27	-6	-2	60	370	9706
27	-6	2	60	370	9706
28	-3	4	105	197	13453
29	1	5	56	67	1517
30	6	7	176	57	11143
31	-3	-1	15	646	1451
31	-3	3	27	261	1460
32	0	5	75	94	3590
33	-4	-1	24	579	3059
33	-4	3	40	263	3090
34	-1	5	96	115	6843
35	-5	-1	35	532	5592
35	-5	3	55	266	5661
36	-2	5	119	133	11532
37	-6	-1	48	499	9289
37	-6	3	72	269	9415
38	-4	0	16	540	1281
38	4	-4	32	182	2561
39	8	4	80	38	6401

Proposition 9.1. *A $D(1)$-tuple with exactly one entry of modulus less than three has at most five elements.*

Proof. The pair $(1, 3)$ has just been excluded. Every other sextuple would yield one of the forty pairs in Table 8, then one of the fifty-six triples in Table 11. The common analytic argument excludes $N(t) \geq N_0$, and the preceding argument excludes $N(t) < N_0$. This contradiction proves the assertion. □

10 Proof of the uniform bound

Proof of Theorem 1.1. If a tuple had at least six elements, choose six and order them by modulus. If none has modulus less than three, Proposition 6.1 gives a contradiction. If exactly one has modulus less than three, Proposition 9.1 gives a contradiction. If at least two have modulus less than three, Proposition 7.1 gives a contradiction. These cases exhaust all sextuples, including ties in modulus, and prove the assertion. □

Funding. The authors were supported by the Croatian Science Foundation under the project no. IP-2022-10-5008 (TEBAG). The authors acknowledge support from the project "Implementation of cutting-edge research and its application as part of the Scientific Center of Excellence for Quantum and Complex Systems, and Representations of Lie Algebras", Grant No. PK.1.1.10.0004, co-financed by the European Union through the European

Regional Development Fund – Competitiveness and Cohesion Programme 2021–2027. This research was funded by the European Union's NextGenerationEU through the National Recovery and Resilience Plan 2021–2026 Institutional grant of University of Zagreb Faculty of Science (IK IA 1.1.3. Impact4Math).

Use of artificial intelligence. The authors used ChatGPT 6 (OpenAI) and Claude Opus 5.5 (Anthropic) to assist with developing proofs, constructing examples and counterexamples, searching the literature, writing code for exact computations, and drafting and revising the manuscript. The authors take full responsibility for the content of this paper.

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